Linked Electron Densities Climb the Banana Ladder: Period Classes of Molecular Integrals over Exponential-Type Orbitals*

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Paper 59 read the radial core of the three-center electron-repulsion integral over $1s$ Slater orbitals as an elliptic Bessel moment: two two-center densities coupled by a Coulomb kernel build a genus-one curve. This paper asks two questions that result leaves open. How does the class climb when an integral links more densities? And how far does the reading reach past $1s$ orbitals? **[SYMBOLIC PROOF + PRIOR ART]** The second is answered first, in part, by a dictionary. Every exponential-type orbital on a center is a polynomial differential operator in its exponent and its center coordinate, with coefficients rational in the exponent, applied to the $1s$ Yukawa function on that center; at the fibre, for s-type densities, every density weight is a polynomial in c, c^{-1} and the Euler operators $c\,\partial_c$ applied to the $1s$ weight. So every multi-density integral with rational kernels is that operator applied to the $1s$ integral of the same density graph and lies in that integral's D-module. For a tree of s-type densities on its one-scale slice the holomorphic period is a differential operator applied to the $1s$ one, so it sits at or below the $1s$ rung, and for $n \leq 8$, when it is not identically zero, on it. **[MEASURED]** In every case computed, all with s-type densities, a general-orbital integral sits on that rung. **[SYMBOLIC + MEASURED]** On the graph the rung is read off the holomorphic period. Any tree of n linked two-center densities has the same period, a Bessel moment in n scales; its one-scale family is annihilated by an operator of order n, which for $n = 2$ to 5 is the Hadamard product of the Legendre operator with the operator of the equal-mass banana integral with $n - 1$ propagators: $n = 2$ is Legendre's K, $n = 3$ Clausen's elliptic square (a K3 surface), $n = 4$ entry #58 of the Calabi–Yau tables, the first period not built from elliptic ones, and $n = 5$ brings no new type. The real three-dimensional star and path of four densities reduce onto that operator with no K3 piece. A fourth distinct center adds no genus **[MEASURED]**. **[SYMBOLIC + MEASURED]** A Coulomb ring of three densities does not climb: its loop integrates out by the three-dimensional uniqueness relation and it sits on the elliptic rung, below the chain of three; the correlated triangle of explicitly correlated methods, with exponential geminals on two edges, sits with the chain on the K3 rung at $\gamma = 1$, 3/2 and 2, its geminal acting as an imaginary external momentum, and its operators follow from integration by parts on the kite and the tetrahedron. A Gaussian geminal leaves the ladder for exponential periods. **[OBSERVATION]** Read as a Feynman graph, every instance fits one pattern: the class is set by the shape of the density graph and the masses on its lines. We state it as a bounded conjecture. Trees sit at rung n; the rung names an operator and not a finite set of numbers, since on every slice tested the three-electron fibre is a K3 function integrated once and at generic scales it has no relation to the K3 and elliptic periods in any basis tried; rings are ordered by uniqueness, not by count; and the integrated value sits at or below its fibre. The caps are stated: the three-dimensional reduction is shown through four densities, the dictionary's rung bound only for s-type trees on the one-scale slice, the only ring tested is the triangle, and the generic fibre's content is evidence. **[PRIOR ART]** The operators are known in the mathematics of Bessel moments and banana integrals; the Averys' momentum-space treatment of these same integrals re-expands every two-center density in a one-center basis and reaches the genus-zero class only through an infinite, non-sparse series. What is new is the route from molecular integrals to the ladder, the dictionary that places every orbital's integral in the D-module of its graph's $1s$ integral, and the prospective records in which most rungs were registered before they were computed.

I. INTRODUCTION

Two-center electron-repulsion integrals over exponential-type orbitals close in elementary functions together with the exponential integral, the logarithm and Euler's constant: a genus-zero class, weight one, settled by Roothaan and Ruedenberg. Paper 59 [1] read the genuine three-center integral $(XY|XZ)$ in momentum space, where each center is a phase and the two two-center densities carry Fock dispersion roots $\Delta_i = \sqrt{c_i k^2 + 1}$ with different scales, and found its radial core on the elliptic curve

$$y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1), \tag{1}$$

whose period at zero separation is a complete elliptic integral,

$$\int_0^\infty \frac{dk}{\sqrt{(c_1 k^2 + 1)(c_2 k^2 + 1)}} = \frac{1}{\sqrt{c_{\max}}}\, K(m), \qquad m = 1 - \frac{c_{\min}}{c_{\max}}, \tag{2}$$

* Paper 63 in the Geometric Vacuum series

degenerating to the elementary $\pi/(2\sqrt{c})$ only at equal scales. That paper's object was one integral. This paper's object is the *ladder*: what happens to the class as an integral links more electron densities, and whether the reading depends on the $1s$ orbitals it was found on.

The answer to the second question is, in part, a dictionary (Sec. II), and it is classical in its parts. Every exponential-type orbital is a derivative of the $1s$ Yukawa function in its exponent and its center; so every integral over such orbitals is a derivative of the $1s$ integral of the same *density graph*, the graph whose lines are the orbitals and the kernels and whose vertices are the electrons and the nuclei. Derivatives in the parameters of a period stay in its D-module. Every such integral therefore lies in the D-module of its graph's $1s$ integral, and on the one-scale slice of a tree of s-type densities its holomorphic period is a differential operator applied to the $1s$ one, at or below its rung and, for $n \le 8$ when it is not identically zero, on it (Proposition 2). The answer to the first question (Sec. III) was found on $1s$ orbitals: for a tree of n linked $1s$ densities the holomorphic period depends only on n, and the operators of the one-scale family for $n = 2, \dots, 5$ are, under one exact Hadamard product, the operators of the equal-mass banana Feynman integrals; for those n the n-density rung is that of the banana carrying a Calabi–Yau $(n-1)$-fold [43]. Every general-orbital case computed, all s-type, sits on the same rung. The Coulomb ring of three does not climb, the correlated triangle does, and a Gaussian geminal leaves the ladder.

The reading that organizes these facts is a Feynman-graph one (Sec. IV): densities and kernels are lines, exponents and geminal decays are masses, and the class is set by the shape of the graph and the masses on its lines. We state it as a bounded conjecture with its falsifiers, and we record the one place its naive form fails: the rung names an operator, not a finite basis of periods. In the transcendence bookkeeping of the Geometric Vacuum program (Paper 18 [44]) this gives the many-electron integrals a second coordinate beside the Tate weight of the one-body and spectral projections: the dimension of the Calabi–Yau variety on whose rung the density graph sits. The seam catalogue (Paper 57 [2]) records the rows; the number-theoretic structure of the first rung is the companion Paper 61 [46].

Most of the mathematics below is prior art, and Sec. V says whose. The route from a molecular integral to it was not found in a targeted search, and most rungs were registered in committed records before they were computed (Sec. III names each record). The caps are stated in Sec. VI.

II. THE DICTIONARY: EVERY EXPONENTIAL-ORBITAL INTEGRAL IS A DERIVATIVE OF A $1s$ YUKAWA INTEGRAL

Setting. An exponential-type orbital on a center $\boldsymbol{A}$ is a finite sum of terms $|\boldsymbol{r}-\boldsymbol{A}|^{j}\, e^{-\zeta|\boldsymbol{r}-\boldsymbol{A}|}\, \mathcal{Y}_{\ell m}(\boldsymbol{r}-\boldsymbol{A})$ with $j \ge 0$ an integer and $\mathcal{Y}_{\ell m}$ a regular solid harmonic (homogeneous of degree ℓ). Slater orbitals are the single terms $j = n-1-\ell$; hydrogenic and Coulomb–Sturmian orbitals, a Laguerre polynomial times an exponential, are such finite sums. The $1s$ seed is the Yukawa function $g_\zeta(s) = e^{-\zeta s}/s$, $s = |\boldsymbol{r}-\boldsymbol{A}|$, the three-dimensional Green's function of $-\nabla^2 + \zeta^2$, whose Fourier transform is $4\pi/(p^2+\zeta^2)$. A *density graph* Γ has a vertex for every electron (integrated) and every nucleus (fixed), a line of mass ζ from a nucleus to an electron for every orbital, and a line between two electrons for every kernel: massless for the Coulomb kernel, mass γ for a Slater geminal $e^{-\gamma r_{12}}$, which is itself $-\partial_\gamma$ of the Yukawa line. The $1s$ *integral of* Γ, $I_\Gamma^{(1s)}(\zeta_e; \boldsymbol{X}_v)$, is the integral with every orbital line the Yukawa g_{ζ_e}.

Lemma 1 (orbital level; prior art)**.** *Every exponential-type orbital on $\boldsymbol{A}$ is $\mathcal{D}(\partial_\zeta, \nabla_{\boldsymbol{A}})\, g_\zeta(|\boldsymbol{r}-\boldsymbol{A}|)$ for a polynomial differential operator $\mathcal{D}$ whose coefficients are rational in ζ.*

[SYMBOLIC PROOF] Two identities do it. Powers are ζ-derivatives, $s^j e^{-\zeta s} = (-\partial_\zeta)^{j+1} g_\zeta(s)$. Solid harmonics are center derivatives: by Hobson's theorem, $\mathcal{Y}_{\ell m}(\nabla) F(s) = \mathcal{Y}_{\ell m}(\boldsymbol{r}-\boldsymbol{A})\,(s^{-1}d/ds)^\ell F(s)$ for any radial F, and $\nabla = -\nabla_{\boldsymbol{A}}$ on a function of $\boldsymbol{r}-\boldsymbol{A}$. Applied to $F = s^{m-1}e^{-\zeta s}$ this gives $\mathcal{Y}_{\ell m}(\boldsymbol{r}-\boldsymbol{A})\, e^{-\zeta s}\sum_{i=0}^{\ell} a_i(m,\ell)\,\zeta^{\,i}\, s^{m-1-2\ell+i}$ with $a_\ell = (-1)^\ell$; the system in m is triangular in the power of s with leading coefficient $(-\zeta)^\ell \neq 0$, so every $\mathcal{Y}_{\ell m}(\boldsymbol{r}-\boldsymbol{A})\, s^j e^{-\zeta s}$ is a combination of the $\mathcal{Y}_{\ell m}(-\nabla_{\boldsymbol{A}})(-\partial_\zeta)^m g_\zeta$ with coefficients in $\mathbb{Q}[\zeta, \zeta^{-1}]$. (For $\ell = 1$: $\mathcal{Y}_{1m} e^{-\zeta s} = -\zeta^{-1}[\,T_2 + \zeta^{-1} T_1\,]$ with $T_m = \mathcal{Y}_{1m}(-\nabla_{\boldsymbol{A}})(-\partial_\zeta)^m g_\zeta$.) In momentum space the same statement reads: the transform of every exponential-type orbital is $e^{-i\boldsymbol{p}\cdot\boldsymbol{A}}\, \mathcal{Y}_{\ell m}(i\boldsymbol{p})\, R(p^2, \zeta)$ with R rational, since $\mathcal{Y}_{\ell m}(i\nabla_{\boldsymbol{p}})$ acting on a rational function of p^2 gives $\mathcal{Y}_{\ell m}(i\boldsymbol{p})$ times a rational function. **[PRIOR ART]** Both halves are classical: the rational momentum-space form of exponential-type orbitals is classical [19], generating angular momentum by differentiating in the nuclear coordinates is standard [19], and the ζ-derivative is the device Paper 59 already uses to restore a Slater orbital from a Yukawa. Guard: `tests/test_paper63_dictionary.py` (the $\ell = 1$ inversion and the $\ell = 2$ triangular system in exact arithmetic; the rational momentum-space form for (n, ℓ) up to $(4, 3)$).

Lemma 2 (integral level)**.** *Let I_Γ be a multi-density integral over exponential-type orbitals with Coulomb or Slater-geminal kernels. Then $I_\Gamma = \prod_e \mathcal{D}_e\, I_\Gamma^{(1s)}$, the operators acting on the exponents and centers of the $1s$ integral of the same graph.*

[SYMBOLIC PROOF] Insert Lemma 1 for each orbital. The $1s$ integrand is bounded by a product of exponentials in the $|\boldsymbol{r}_i - \boldsymbol{X}_v|$ times integrable Coulomb singularities, uniformly for exponents in a compact subset of $(0, \infty)$ and centers in a compact set, as are its derivatives of every order in those parameters; dominated convergence moves the $\mathcal{D}_e$ outside the integral. The same holds

for the momentum form, where the orbital lines are rational and the exponential decay of position space becomes the polynomial decay of the integrand in momentum.

Proposition 1 (every orbital lies in the $1s$ module)**.** *Let $\mathcal{M}$ be the holonomic D-module generated by $I_\Gamma^{(1s)}$ over the exponents $\zeta > 0$ and the centers. Every I_Γ over exponential-type orbitals lies in $\mathcal{M}$. At the fibre, where the centers coincide, a density built from s-type orbitals enters through its dispersion root $\Delta_v = \sqrt{c_v L_v^2 + 1}$ alone, and every such weight $P(L_v^2)\,\Delta_v^{-(2m+1)}$ with $\deg P \le m$ is a polynomial in c_v, c_v^{-1} and the Euler operator $c_v\partial_{c_v}$ applied to Δ_v^{-1} (for $\ell > 0$ the center derivatives of Lemma 1 leave polynomial momentum factors at the fibre, which only the module statement covers):*

$$\Delta^{-(2m+1)} = \Big(1+\tfrac{2}{2m-1}\,c\,\partial_c\Big)\Delta^{-(2m-1)}, \qquad L^2\,\Delta^{-j} = c^{-1}\big(\Delta^{-(j-2)} - \Delta^{-j}\big). \tag{3}$$

[SYMBOLIC PROOF] The first statement is Lemma 2 read in $\mathcal{M}$: a section of a holonomic module and its derivatives generate a submodule. For the fibre identities, $\partial_c\Delta^{-1} = -\frac{1}{2}L^2\Delta^{-3}$ and $L^2 = (\Delta^2-1)/c$ give $c\,\partial_c\Delta^{-1} = -\frac{1}{2}(\Delta^{-1}-\Delta^{-3})$, which is the case $m=1$; induction on m gives the rest, and the second identity is $L^2 = (\Delta^2-1)/c$ again. **[MEASURED]** On the two-density rung the identities are checked against the period (2): $\int_0^\infty \Delta_1^{-3}\Delta_2^{-3}dk$ and $\int_0^\infty \Delta_1^{-5}\Delta_2^{-5}dk$ equal the operators of Eq. (3) applied to $K(m)/\sqrt{c_{\max}}$, to 10^{-25} at three scale pairs, and the result is again a combination of K and E: the Legendre rung. On the three-density rung the Euler operators act under the tree identity (Eq. (5) below): the one-dimensional transform of the $1s$ weight is $2c^{-1/2}K_0(x/\sqrt{c})$, and Eq. (3) turns it into $2xc^{-1}K_1(x/\sqrt{c})$ for Δ^{-3} and $\frac{4}{3}xc^{-1}K_1 + \frac{2}{3}x^2c^{-3/2}K_0$ for Δ^{-5}, the known transforms of those powers. The two-dimensional fingerprint integral of the chain of three, with Δ^{-3} or Δ^{-5} on every density, equals $\pi^{-1}\int_0^\infty \prod_v w_v(x)\,dx$ with those weights to 2×10^{-9} at three scale triples, the precision of the two-dimensional quadrature; the $1s$ weight in place of w_v misses by more than a percent. Guard: `tests/test_paper63_dictionary.py` (29 checks, fire-tested).

What the dictionary does and does not say. It places every general-orbital integral in the D-module of the $1s$ integral of its graph. That alone does not bound the rung: the rung is read on a one-parameter fibre at special values of the other parameters, coincident centers and fixed scales, where derivatives in those parameters need not stay in the one-parameter module of the restricted $1s$ function. For a tree of s-type densities on its one-scale slice the next proposition bounds the holomorphic period, not the fibre, and on that slice proves attainment for $n \le 8$ when the period is not identically zero; for rings, for $\ell > 0$, on other slices and for the fibre itself neither the bound nor attainment is shown.

Proposition 2 (the tree bound)**.** *On the slice $(t;1,\dots,1)$ of a tree of n linked s-type densities, the holomorphic period of the integral is $E\,F_n$, with $F_n(t) = \int_0^\infty K_0(tx)K_0(x)^{n-1}dx$ and E a differential operator in t with Laurent-polynomial coefficients. Its composition factors are among those of the operator of F_n, so it sits at or below rung n. If $n \le 8$ and the period is not identically zero, as it is not when every weight is a pure power $\Delta_v^{-(2m+1)}$, its minimal operator is equivalent to that of F_n: it sits on rung n.*

[SYMBOLIC PROOF] The derivation of the tree identity, Eq. (5) below, holds for any radial density weight, with each weight's one-dimensional transform in place of $K_0(a_vx)$; by Eq. (3) the s-type transforms at unit scale are combinations, with coefficients polynomial in x^2, of $K_0(x)$ and $xK_1(x)$. With $\theta = x\,d/dx$, $\theta K_0 = -xK_1$ and $\theta(xK_1) = -x^2K_0$, so

$$\theta\big(x^{2k}K_0^{a+1}(xK_1)^{b-1}\big) = 2k\,x^{2k}K_0^{a+1}(xK_1)^{b-1} - (a+1)\,x^{2k}K_0^{a}(xK_1)^{b} \tag{4}$$

and recursion on b writes every product $x^{2k}K_0^a(xK_1)^b$, $a+b = n-1$, as $Q(x^2,\theta)K_0^{n-1}$, dividing only by $a+1 \le n-1$. Against $K_0(tx)$, integration by parts turns θ into $-(1+\theta_t)$, the boundary terms vanishing (logarithmic at 0 times x, exponential at infinity), and $x^2K_0(tx) = t^{-2}(\theta^2K_0)(tx) = t^{-2}\theta_t^2K_0(tx)$; the scaled density's weights are images of $K_0(tx)$ under θ_t, since $(tx)K_1(tx) = -\theta_tK_0(tx)$. So the period is $E\,F_n$, and the module it generates is a submodule of that of F_n. For $n = 2$ to 8 the operator of F_n is irreducible (Sec. III C; guards `tests/test_paper63_four_density_rung.py`), so that module is simple: a period that is not identically zero generates all of it, and its minimal operator is equivalent to that of F_n. The one-dimensional transform of $\Delta^{-(2m+1)}$ is a positive multiple of $x^mK_m(x/\sqrt{c})$, so with pure-power weights the period is positive. **[MEASURED]** For $n = 3$ and 4 and nine products, with operators of up to six terms, the identity holds at three values of t each to 4×10^{-38}, against a quarter to four fifths for a recursion with one wrong denominator (record `docs/predictions/2026-11-06_rung_bound_trees.md`, registered before computing; guard `tests/test_paper63_tree_bound.py`).

In every case computed the general-orbital integral sits on the $1s$ rung: the chain at the Slater weights has the same composition factors as the chain at the fingerprint weights (Sec. III D). Two caps it inherits unchanged from Paper 59: the rung is read on the fibre and its holomorphic period, and the integrated value, over the Feynman parameters and separations, may sit lower; and the reduction of the real three-dimensional integral to the one-dimensional Bessel-moment family, which the dictionary does not supply, is shown below for the chain of three and the two trees of four only. Gaussian orbitals and Gaussian geminals are outside the lemma: $e^{-\alpha r^2}$ is no derivative of a Yukawa, its transform is not rational, and Sec. III D finds its integrals off the ladder.

III. THE RUNGS: A FOURTH CENTER, A THREE-ELECTRON CHAIN, FOUR LINKED DENSITIES, AND A RING

Two results pre-registered on 2026-10-13 (records `docs/predictions/2026-10-13_four_centre_genus.md` and `docs/predictions/2026-10-13_three_electron_k3.md`, each committed before its computation) extend the genus reading of Paper 59 [1] in two directions: more centers, and more Coulomb kernels. A third, registered on 2026-10-16 (`docs/predictions/2026-10-16_linking_ceiling.md`, written before its computation but not committed beforehand), asks how the rung grows with the number of linked densities. A fourth, registered and committed on 2026-10-17 before its computation (`docs/predictions/2026-10-17_density_loop.md`), asks whether a ring of linked densities climbs faster than a tree. A fifth, committed the same day before its computation (`docs/predictions/2026-10-17_ring_extensions.md`), asks whether the correlated triangle, with geminals on two edges, behaves like the Coulomb one, and how a ring of four behaves; the second test stopped for feasibility. A sixth, committed before its computation (`docs/predictions/2026-10-17_ising_class.md`), asks where the Ising-class moments sit on the ladder.

A. A fourth distinct center adds no genus

[MEASURED] The general four-center integral $(XY|ZW)$, four distinct centers and $1s$ Slater orbitals, reduces by the same Feynman parametrization to one radial k-integral over two parameters (s,t) whose integrand carries exactly the two dispersion roots $\Delta_1(s)$, $\Delta_2(t)$ of the curve (1); the centers enter only through a single phase vector $\boldsymbol{W}(s,t)$ and the two pair distances D_1, D_2. The reduced formula gives 0.218814149639 against 0.218814149425 from McMurchie–Davidson on sixteen-Gaussian fits (the residual is the fit gap), and reproduces the three-center value 0.204941652531 of Paper 59 as a special case. There is no genus two at four centers, consistent with the class being set by the number of distinct two-center densities the kernel couples, not by the number of nuclei; this is that rule's first instance beyond the three-center integral of Paper 59. Its honest weight is that of a corollary of Paper 59's reduction, a calibration rather than a hard test. The momentum form itself is the Averys' (Ref. [5], App. B); the genus reading is not in it. Guards: `tests/test_paper63_four_centre_genus.py` (both values recomputed against McMurchie–Davidson; the kernel derived symbolically as the second z-derivative of a Yukawa product; the integrand's square roots and the dependence on the pair placement checked).

B. A three-electron chain lives on a K3 surface

[SYMBOLIC + MEASURED] The chain integral $\langle\rho_a(1)\rho_b(2)\rho_c(3)/(r_{12}r_{13})\rangle$, three two-center $1s$ Slater transition densities (dispersion roots $\Delta_b = \sqrt{c_b p^2+1}$, $\Delta_c = \sqrt{c_c q^2+1}$, $\Delta_a = \sqrt{c_a|p+q|^2+1}$, scales $a = c_a^{-1/2}$, $b = c_b^{-1/2}$, $c = c_c^{-1/2}$) joined by two Coulomb kernels at electron 1, has two momentum integrations and three dispersion roots. The angular integral is elementary (no logarithm); the reduced two-dimensional fibre matches the direct three-dimensional one (10^{-15} to 4×10^{-14} in the record, 10^{-9} in the reduced-precision guard); the one-center limit is rational (23/54, 133/144, 289/225 at the three registered exponent triples); the branch locus in $\mathbb{P}^2$ is six lines with no four concurrent, so the double cover is a K3 surface, and fixed-p slices are non-isotrivial genus-one curves. **[MEASURED + PRIOR ART]** The holomorphic period reduces by one Fourier transform to the three-scale Bessel moment $\bar{V}_3 = \int_0^\infty K_0(at)K_0(bt)K_0(ct)\,dt$ (the notation of Ref. [22]), whose closed form for $c \geq b \geq a > 0$ is $(2c/\pi)\,\bar{V}_3 = K(k_-)K(k'_+) + K(k_+)K(k'_-)$, with the moduli $k_\pm$ of Ref. [22] (Eq. 33, obtained there from an identity of Bailey [23]), checked at four scale triples to 10^{-34} in the record and at three in the guard to 10^{-17}. The K3's period is therefore "elliptic squared": the holomorphic period opens no new transcendence class, being a sum of products of periods of the genus-one class of Paper 59. The reduction from a Slater chain to $\bar{V}_3$ is this paper's; the closed form is that of Ref. [22]. In the language of Paper 59's literature placement, the repulsion integral sits on the rung of the two-loop sunrise and the chain on the rung of the three-loop banana; the families differ, and Sec. III C gives the exact relation. **[OBSERVATION, evidence]** At equal scales the true fibre at zero separation collapses to two K3 periods (Bessel moments of the record) and π^3. At four generic scale triples, $(1;2,4)$, $(1;3,7)$, $(3;1,7)$ and $(3;4,6)$ with the central density's scale first, integer-relation searches at 340 and 400 digits find no relation in five bases built from the K3 periods and their scale derivatives, products of elliptic periods at the Bailey moduli, residue periods and their products, dilogarithms and trilogarithms of scale ratios, the vacuum sunrise and its mass derivatives, logarithms times periods, and weight-three Tate numbers. Every basis recovers a planted relation, and in every basis the fibre with $\int_0^x K_0$ dropped from each leaf reduces onto the K3 periods; three of the twenty searches are void, their precision cap below the planted relation's height. Moving the central density within the scale set $\{1,3,7\}$ gives no relation either, and the fibre does not collapse at the two threshold triples $(2;1,1)$ and $(3;1,2)$, where the central scale is the sum of the others. That is evidence, at a sample of triples and with guessed bases, that the generic fibre carries content beyond the K3 and elliptic periods, as the banana does; on a one-parameter slice it is a K3 function integrated

once (Sec. III D), a class no finite basis of periods is expected to capture. **[SYMBOLIC + MEASURED]** A second slice, with scales $(t;1,2)$, passes through two of these triples, $(1;2,4)$ at $t=1/2$ and the threshold triple $(3;1,2)$ at $t=3$, where the fibre values agree with the slice to 20 digits. Its Mellin operator has order 17 and divides exactly over the factor $\theta+2$ and two first-order pieces, one for each leaf, which enter together as their least common left multiple. Above them sits a piece of order four, equivalent by an exact map with poles only at $t=0,\pm1,\pm3$ to the operator of the three-scale moment $\int_0^\infty K_0(tx)K_0(x)K_0(2x)\,dx$. The composite of order seven annihilates the fibre to 5×10^{-33} of its largest term at 32 digits, the fibre needs the order-four piece, and a search modulo a prime finds no rational map splitting it off within the bound set by the local exponents, while the same call finds a planted split. So on this slice too the fibre is an elementary function plus a K3 function integrated once, which accounts for the null at those two triples (record `docs/predictions/2026-10-21_generic_slice.md`). Longer trees are taken up in Sec. III C. Guards: `tests/test_paper63_three_electron_k3.py` (the fibre agreement, the elementary angle integral, the rational one-center limits, the closed form against quadrature, the K3 period against $\bar V_3$, the six-line branch locus, the non-isotrivial slice); `tests/test_paper63_generic_slice.py` (the second slice); the equal-scale collapse and the generic null are not guarded, and the records carry them (`docs/predictions/2026-10-13_three_electron_k3.md`, `docs/predictions/2026-10-19_p3e2_widened.md`).

C. Four linked densities: the first rung beyond elliptic

[SYMBOLIC + MEASURED] The holomorphic period of a tree of linked densities depends only on how many densities the kernels link, not on the pattern of the links; a Coulomb ring of three behaves differently (Sec. III D). Take n two-center $1s$ densities joined into a tree by $n-1$ rational kernels, and write the dispersion root of density v at the fingerprint fibre as $\Delta_v=\sqrt{c_vL_v^2+1}$, with L_v the signed sum of the momenta on the kernels at v. Fourier-transforming each $1/\Delta_v$ to $K_0(|y|/\sqrt{c_v})/(\pi\sqrt{c_v})$ turns every momentum integration into a delta function that identifies the two ends of its kernel, and a tree is connected, so

$$\int_{\mathbb{R}^{n-1}} d^{n-1}k\,\prod_{v=1}^{n}\frac{1}{\Delta_v(L_v(k))}=\frac{2^n}{\pi\sqrt{c_1\cdots c_n}}\int_0^\infty\prod_{v=1}^{n}K_0(a_vx)\,dx,\qquad a_v=\frac{1}{c_v},\tag{5}$$

for a star and a chain alike. At $n=3$ this is the $\bar V_3$ reduction of Sec. III B; at $n=2$ it is the elliptic period of Eq. (2). A three-dimensional evaluation of four-density stars and chains, with the faster-decaying roots Δ_v^{-3} to which the same argument applies, matches the Bessel-moment side to between 4×10^{-9} and 2×10^{-7}, the grid's own convergence. That check tests the identity's bookkeeping; the real three-dimensional integral reduces to moments of the same family with other weights in x, as shown below for both trees of four densities, the star and the path.

[SYMBOLIC] To read the rung, vary one scale: $F_n(t)=\int_0^\infty K_0(x)^{n-1}K_0(tx)\,dx$. The Mellin transform of K_0^{n-1} obeys the recurrence of the $(n-1)$-th symmetric power of the Bessel operator, and that of $K_0(tx)$ is $2^{s-2}\Gamma(s/2)^2$, so F_n satisfies an explicit linear equation in t of order n; **[MEASURED]** each one is checked against direct quadrature of F_n to 10^{-28}. For $n=2$ it is the Legendre equation of K. For $n=3$ it is the equation of ${}_3F_2(\frac12,\frac12,\frac12;1,1;t^2/4)$, Clausen's square of an elliptic period, as Bailey's formula requires. For $n=4$, in the variable $x=1/(4t^2)$, it is

$$\theta^4-x(2\theta+1)^2(10\theta^2+10\theta+3)+9x^2(2\theta+1)^2(2\theta+3)^2,\qquad \theta=x\,\frac{d}{dx},\tag{6}$$

which has maximally unipotent monodromy at $x=0$ and conifold points at $x=1/36$ and $x=1/4$, with local exponents $(0,1,1,2)$. Its holomorphic solution is $\sum_m B_m(x/4)^m$ with integer $B_m=\binom{2m}{m}^2\sum_k\binom{m}{k}^2\binom{2k}{k}$, the term-by-term (Hadamard) product of $\binom{2m}{m}^2$ with the Apéry-like sequence of the equal-mass sunrise, and its mirror map has integer coefficients.

[SYMBOLIC] Whether a family is built from elliptic periods shows in the quadratic relations among the solutions of its equation: for monodromy of these types, an elliptic family has none, the square of one elliptic family or the tensor product of two has one, and the cube of one has three. Counted exactly on Taylor bases at two rational points (one in the guard), the numbers are $0,1,0,1$ for $n=2,3,4,5$. For $n=4$ the operator is irreducible, with no bound on degrees: a right factor of order k would carry, at each singular point, k of the operator's local exponents, and Fuchs' relation then fails for every $k<4$, however many apparent singularities the factor adds. The same count shows the operators for $n=2,3,5$ irreducible. Its Yukawa coupling, $1+16q+1152q^2+99520q^3+\cdots$, is not constant, as it would be for a symmetric cube. An irreducible fourth-order local system with no invariant quadric that is not a symmetric cube has differential Galois group Sp_4 or SL_4, and neither is a quotient of a product of copies of SL_2. So F_4 is not a polynomial in elliptic periods with algebraic moduli, and neither is the generic four-scale moment, which restricts to it. Two linked densities give an elliptic period, three give a K3 whose period is elliptic squared, and four give the first period of a new kind. At $n=5$ the single relation marks an orthogonal structure of rank five, not the fourth power of an elliptic period, which would carry six. That is the structure of the exterior square of a fourth-order Calabi–Yau operator, and the tables of Ref. [29] print the operator of F_5 verbatim, attached to entry #39, a different operator from the

#58 of $n = 4$. So five linked densities give no new type of period: the type is that of four, in a different family.

[**OBSERVATION**] Every electron-repulsion integral couples exactly two densities, so over s-type orbitals the integrals of a molecular Hamiltonian stay on the elliptic rung however large the molecule (for $\ell > 0$ this is not shown). The class climbs only in correlated integrals that link more densities, such as the four-electron bridge integrals of explicitly correlated methods taken over two-center densities. Two caps apply. These are statements about the fibre and its holomorphic period, so the integrated value may sit lower, as the two-center exchange class does (Paper 59, geminal section). And the reduction of the real three-dimensional integral into the family of Eq. (5) is shown only for the chain of three and the two trees of four below; for five or more densities it is not shown. [**SYMBOLIC + MEASURED**] At the fingerprint fibre on the slice $(t; 1, 1, 1)$ the star is $(16/\pi^4)\, t^3 \int_0^\infty \chi(x)^3 K_0(tx)\, dx/x$, with $\chi = \int_0^x sK_1(s)\, ds$ (Sec. III D), checked against a momentum-space evaluation to 2×10^{-21}. Its Mellin operator, of order 19, divides exactly with a piece of order four equivalent to the operator (6) by an exact rational map. An order-eight right factor annihilates the star to 3×10^{-45} of its largest term at 45 digits; its composition factors are that piece, the Legendre operator of F_2 and two of order one, with no K3 piece, and the star needs the top piece. So the star sits on the rung of four linked densities and skips the K3 rung; minimality is bounded by a search modulo a prime, not proved (record `docs/predictions/2026-10-21_four_density_star.md`). Guards: `tests/test_paper63_four_density_rung.py`, `tests/test_paper63_four_density_star.py`. [**SYMBOLIC + MEASURED**] The path, four densities in a chain, reduces the same way. On the slice $(1, t, 1, 1)$, with t on an inner density, it is $(16/\pi^4)\, t^3 \int_0^\infty \int_0^\infty \min(x, y)\, K_0(tx)\chi(x)\, K_0(y)\chi(y)\, dx\, dy$, checked against a momentum-space evaluation to 20 digits. Its holonomic relations give an operator of order 74 with an exact right factor of order 12, which annihilates the path to below 10^{-28} of its largest term at 32 digits and 4×10^{-56} at 60. That factor splits exactly into a piece equivalent to the operator (6), three pieces equivalent to the Legendre operator of F_2, and two of order one, each equivalence by an exact rational map; there is no K3 piece, and the path needs the top piece. So both trees of four densities sit on the same rung and skip the K3 rung; for the path, minimality is bounded by a search modulo a prime up to t-degree 260 (record `docs/predictions/2026-10-22_four_density_path.md`). Guard: `tests/test_paper63_four_density_path.py`.

[**PRIOR ART**] Most of this structure is known in the mathematics of Bessel moments; what is new here is the route into it from a molecular integral. The operator (6) is entry #58 of the Calabi–Yau tables of Almkvist, van Enckevort, van Straten and Zudilin [29] (entry 2.9 of the Calabi–Yau operator database [30]), listed there as a Hadamard product, with the same holomorphic coefficients; Gegelia and van Straten [31] list it among their conjectural sources of paramodular forms, at conductors 2145 and 2465 (their Table 6; their Table 3 lists entry 2.9 at conductors 945 and 960). For every n from 2 to 5 the holomorphic solution of the equation of F_n at $t = \infty$ is $t^{-1} \sum_m \binom{2m}{m}^2 W_{n-1}(2m)\, (16t^2)^{-m}$, the Hadamard product of $\binom{2m}{m}^2$ with the even moments $W_{n-1}(2m)$ of the $(n-1)$-step planar random walk. The Hadamard construction, and the tables that place its $n = 4$ and $n = 5$ cases at entries #58 and #39, are Almkvist and Zudilin's [33]; the links of W_N to Bessel moments, random walks and Calabi–Yau equations are Broadhurst's [40]; Vanhove's operators for the equal-mass two-dimensional banana integral with N propagators [34] annihilate $\sum_k W_N(2k) s^{-k-1}$, checked here in exact arithmetic for $N = 2$ to 5, so that series is the banana's holomorphic period at large s. The alternation of the monodromy, symplectic for n even and orthogonal for n odd, matches the self-duality of the symmetric-power Kloosterman motives of Fresán, Sabbah and Yu [26], whose pairing is symplectic when the number of Bessel factors is even and orthogonal when it is odd. [**SYMBOLIC**] So the tree ladder is tied to the banana ladder, where an l-loop equal-mass banana integral carries a Calabi–Yau $(l-1)$-fold [43], by one exact operation: with $y = 1/(16t^2)$ and $F_n = t^{-1} g(y)$, the operator of g is the Hadamard product of the Legendre operator with the operator of the banana with $n-1$ propagators, checked in exact arithmetic for $n = 2$ to 5. Since $\int_0^\infty J_0(px)\, dp = 1/x$, the same fact reads $F_n(t) = \int_0^\infty dp \int_0^\infty x\, J_0(px)\, K_0(tx) K_0(x)^{n-1} dx$: F_n is the n-propagator banana, with masses $(t, 1, \ldots, 1)$, integrated over a line of external momentum. It sits on the rung of the banana with $n+1$ propagators but is not that family: its singular values of t^2, $\{1\}, \{4\}, \{1, 9\}, \{4, 16\}$ for $n = 2, \ldots, 5$, are the thresholds of the banana with $n-1$ propagators.

[**SYMBOLIC + MEASURED**] At $t = 1$ the ladder is the Ising class of Bessel moments, $F_n(1) = c_{n,0} = \int_0^\infty K_0(x)^n\, dx$ [22]. The singular values of t^2 are the thresholds $(n-1-2j)^2$ for every n up to 8, with no other finite nonzero singular point, so $t = 1$ is a singular point of F_n's operator exactly when n is even; the operators for $n = 6$, 7 and 8 are irreducible by the exponent argument above, so these are the singular points of F_n's minimal operator. There one local exponent is doubled, while at the thresholds of odd n one is a half-integer. Since F_n is analytic for $t > 0$, at a threshold it lies among the solutions without the logarithm or the branch. A prediction registered and committed before the computation (`docs/predictions/2026-10-17_ising_class.md`) read this as a rule for values: odd moments stay on their rung and even ones drop. The registered value cases hold, with moments for $n = 2$ to 8 to 286 digits by two routes and every informative integer-relation search controlled; the registered attachment of $c_{4,0}$ to the conifold fibre was not tested. $c_{3,0}$ is a complex-

multiplication elliptic value; $c_{5,0}$ and $c_{7,0}$ fit no tested lower basis, at coefficient bounds of 10^{14} to 10^{17} and 10^{7} to 10^{9}; $c_{4,0} = 4\pi^2 L(f,2)$ with $f = \eta(2\tau)^4\eta(4\tau)^4$ of weight four and level eight, checked to 5×10^{-261}; and $c_{6,0} = 108\pi^2 L(g,4)$ with g of weight six and level 24, to the same precision (both identities guarded to 10^{-25} in `tests/test_paper63_four_density_rung.py`; the 5×10^{-261} precision is the record's). But $c_{8,0}$ fits no single L-value tried, with no Tate relation to coefficient bound 10^7 and none with an L-value to 10^{12}, so the $n=8$ search gives parity no support as a rule for values. **[PRIOR ART]** The L-value forms of $c_{3,0}$ and $c_{4,0}$ are collected by Lim, Tu and Yu [36], who credit them to earlier work, through the hypergeometric forms of Ref. [22]; the numerical form of $c_{6,0}$ is theirs, and in their account of the underlying Kloosterman motives the moments at $n=5$, 7 and 8 enter only through 2×2 determinants of periods. The singular points themselves are the thresholds and pseudo-thresholds of the equal-mass banana operator (Ref. [43], Eq. (3.7)); the threshold reading of the values is what the ladder adds. In the bookkeeping of Papers 18 and 34 [44, 45] these values lie beyond the genus-graded embedding tier: $c_{3,0}$ is a rational multiple of $2^{-2/3}\Gamma(1/3)^6/\pi$, a complex-multiplication period, and $c_{4,0}$ and, numerically, $c_{6,0}$ are π^2 times critical L-values of modular forms, periods of the Kloosterman motives. The nearest physics analogue of a ceiling set by the calculation rather than the system is the bound Bourjaily, McLeod, von Hippel and Wilhelm argue for on the rigidity of massless four-dimensional Feynman integrals, $2(L-1)$ at L loops [41]: there the loop order caps the geometry, here the number of densities one integral links. On the chemistry side, one-center three- and four-electron correlated integrals have closed forms [6, 42], consistent with the rung needing two-center densities. A targeted search found no source placing a molecular integral on these operators; that is an absence from a search, not a proof of absence. Matuzas [32] relates one channel of a different four-electron correlation integral, a ring of Coulomb links with one diagonal over Slater functions, to another Calabi–Yau operator, AESZ-330, by an inhomogeneous equation.

D. A ring of three densities: the loop does not climb

Every rung above is a tree. An integral sits on a rung when the minimal annihilator of its fibre has no composition factor above that rung. A ring of kernels leaves one momentum that the delta functions behind Eq. (5) do not remove, and one might expect a loop to raise the class faster than a leg does. The test case is the Coulomb triangle $T = \langle \rho_a(1)\rho_b(2)\rho_c(3)/(r_{12}r_{13}r_{23})\rangle$, three densities with every pair joined by a kernel. Its explicitly correlated relative $\langle f_{ik}f_{jk}/r_{ij}\rangle$ is the one three-body class of the LiH correlated engine that does not separate. The prediction, written and committed on 2026-10-17 before the computation (`docs/predictions/2026-10-17_density_loop.md`), was that the loop does not climb.

[SYMBOLIC + MEASURED] The loop integrates out exactly. In three dimensions three massless propagators of unit index are unique, and the uniqueness (star-triangle) relation gives

$$\int \frac{d^3l}{|l-x_1|^2\,|l-x_2|^2\,|l-x_3|^2} = \frac{\pi^3}{|x_1-x_2|\,|x_2-x_3|\,|x_3-x_1|}, \tag{7}$$

which a Feynman-parameter evaluation confirms at five random triples to 1.2×10^{-12}. The triangle becomes an integral over the same two momenta and three densities as the three-electron chain. For spherical densities the magnitude measure cancels the weight, and

$$T = \frac{1}{\pi}\int_{|p-q|\le w\le p+q} \tilde\rho_a(p)\,\tilde\rho_b(q)\,\tilde\rho_c(w)\,dp\,dq\,dw. \tag{8}$$

[SYMBOLIC + MEASURED] At one center, with $1s$ Slater densities, Eq. (8) reproduces Remiddi's closed form [7], as written in Ref. [21] and differentiated in the exponents, to 2×10^{-21}. Where the chain is rational (Sec. III B), the triangle carries a dilogarithm. At exponents $(1,1,1)$

$$T = \frac{11}{3} - \frac{\pi^2}{8} - \frac{3}{2}\,\mathrm{Li}_2\!\left(-\tfrac12\right) - \frac{46}{9}\ln 2 - \frac{3}{4}\ln^2 2 + \frac{3}{2}\ln 2\ln 3, \tag{9}$$

and at $(1,2,3)$ a 100-digit integer-relation search finds a closed form in $\mathrm{Li}_2(1/3)$, $\mathrm{Li}_2(1/5)$, $\mathrm{Li}_2(1/6)$, π^2 and products of logarithms. The ring raises the weight from zero to two. In the bookkeeping of Papers 18 and 34 these one-center values are embedding-tier constants of weight two, one step above the weight-one two-center seed set; the π^3 of Eq. (7) is the loop's normalization.

[SYMBOLIC + MEASURED] At the fingerprint fibre, $\tilde\rho_v = \Delta_v^{-3}$ with scales $(1,1,t)$, the triangle is a moment of three Struve functions,

$$G(t) = \frac{\pi}{2}\,t^2\int_0^\infty x^2\,\psi(x)^2\,\psi(tx)\,dx, \qquad \psi = \mathbf{L}_{-1} - I_1, \tag{10}$$

where ψ is one density with its kernel in position space and solves $(\theta^2 - 1 - x^2)\psi = -2/\pi$. The Mellin route of Sec. III C gives

$$L_G = -4\theta(\theta-3)^2 + t^2\,(5\theta^3 - 7\theta^2 - 5\theta + 3) - t^4\,\theta(\theta+2)(\theta+3), \qquad L_G\,G = \tag{11}$$

with $\theta = t\,d/dt$, checked by quadrature of the Struve form at six values of t to 18 digits, and independently on the momentum form, with the t-derivatives taken analytically. The source term is the operator applied to a constant: $L_G\,1 = 3t^2$, so $L_G(G+1) = 0$, and the constant is what the regularized Mellin transform cannot see. This operator factors. The solution of L_G regular at $t=0$ is algebraic, $(1-t^2/4)^{-3/2}$. Write $L_G = M\,(\theta - \lambda_G)$ with $\lambda_G = 3t^2/(4-t^2)$; the division is exact. The quotient M

is equivalent to the Legendre equation $\theta^2-t^2(\theta+1)^2$ of F_2, the two-density period: the map $f \mapsto t^3(1+\theta)f/(4-t^2)$ sends every solution of the Legendre equation to a solution of M, an identity checked in exact rational arithmetic. And G needs the elliptic piece: $(\theta-\lambda_G)(G+1)$ is a nonzero solution of M, equal to -0.161 at $t=0.5$ and -2.33 at $t=1.3$. Neither side has a relation to the K3 family of F_3. So G is built from an elementary constant, an algebraic function and an elliptic period integrated once against an algebraic kernel. The triangle sits on the elliptic rung of a single repulsion integral. That is one rung below the three-density chain.

[SYMBOLIC + MEASURED] The chain $C(t)$ at the same fibre, treated the same way, is the control. The Mellin route gives it an operator of order seven, which divides exactly as $N_2\,R_3\,R_1\,(\theta-2)$. The two lowest pieces are elementary: $\theta-2$ has the solution t^2, and $R_1=(\theta-3)-t^2(\theta+1)$ has the solution $t^3/(1-t^2)^2$. The piece R_3, of order three, is equivalent to the operator $4\theta^3-t^2(\theta+1)^3$ of F_3: the map

$$f \;\mapsto\; \frac{t^3(t^4-18t^2-4)\,f+t^3(2t^4-43t^2-40)\,\theta f+t(t^6-27t^4-96t^2+32)\,\theta^2 f}{(t^2-4)^2} \tag{12}$$

sends every solution of F_3's operator to a solution of R_3, an identity checked in exact rational arithmetic. The top piece N_2, of order two, is not needed: $R_3R_1(\theta-2)$ already annihilates the chain, with a residual of at most 2×10^{-31} at 32 digits and 4×10^{-19} at 20 digits at four values of t between 0.5 and 3.5, while changing one coefficient of R_3 gives residuals from 1.9 to 604. The chain's only piece beyond the elementary ones is therefore the K3 family of F_3. Finer still, R_3R_1 splits exactly, so the order-five operator is a left multiple of an operator P of order four, and the chain satisfies P: PC, divided by the solution $t^3/(1-t^2)^2$ of R_1, is at most 10^{-23} at 20 digits and 10^{-33} at 30 digits at four values of t, and 9×10^{-45} at 40 digits at one, while a 1% change in one coefficient gives 0.02 to 0.07; the next paragraph derives this. The second solution of $R_1(\theta-2)$, which carries $\operatorname{artanh} t$, is absent. So $C=c\,t^2+t^2\int^t s^{-3}(B'f)(s)\,ds$ with f a solution of F_3's operator and B' a rational map. That integral cannot be done inside the K3 family: no rational map splits the K3 piece off t^2. This is proved by bounding the poles and degree of any such map from the local exponents and searching that bound exhaustively in exact arithmetic, a search that finds every planted split. Because F_3's operator is irreducible, the chain is therefore not a rational combination of F_3 and its derivatives plus any function without a K3 piece: it is a K3 function integrated once against an elementary kernel. So the Coulomb ring lowers the rung at the fibre and raises the weight at one center. In neither limit does it climb the period ladder.

[SYMBOLIC + MEASURED] The same holds for the physical weights. With the Slater densities of Sec. III B, the chain's fibre at zero separation, on the slice with scales $(t;1,1)$ and t on the central density, is

$$G(t)=\int_0^\infty x\,K_1(tx)\,J(x)^2\,dx, \qquad J=3\int_0^x K_0-3xK_0-x^2K_1. \tag{13}$$

The Mellin route gives an operator of order seven, which divides exactly as $N_2\,R_3\,R_1\,(\theta+2)$. The two lowest pieces are elementary: $\theta+2$ has the solution t^{-2}, and $R_1=(1-t^2)\theta-(1+7t^2)$ has the solution $t/(1-t^2)^4$. The piece R_3, of order three, is equivalent to F_3's operator $4\theta^3-t^2(\theta+1)^3$: an explicit order-two map, rational in t with poles at $t=0,\pm2$, sends every solution of F_3's operator to a solution of N_2R_3, and R_3 is the annihilator of the image, both checked in exact rational arithmetic. Again N_2 is not needed: $R_3R_1(\theta+2)$ annihilates G with a residual of at most 2×10^{-31} at 32 digits at five values of t between 0.5 and 3.5, while changing one coefficient of R_3 by one part in 198 gives residuals from 10^3 to 4×10^4. And G needs the K3 piece: $R_1(\theta+2)G$ is nonzero, 191 at $t=0.5$ and 8.9 at $t=3.5$. A search modulo a prime finds no rational map that splits the K3 piece off, in a box larger than the bound set by the leading terms of the local exponents, and the same call finds a planted split. So G is an elementary function plus a K3 function integrated once against an elementary kernel. The second solution of $R_1(\theta+2)$, which carries $\operatorname{artanh} t$, is absent, as it is for the chain. In $w=t^2$, $\delta=w\,d/dw$, the exact system $R_1S=1$ modulo R_3 has a unique rational solution $S=s_0+s_1\delta+s_2\delta^2$ inside the bound set by the local exponents, with denominator $w(w-1)^4(2w+7)$. So $R_3R_1(\theta+2)=Q_1P$ with $P=(\theta+2)-S\,R_1(\theta+2)$ of order four, and P kills t^{-2}. On G, P leaves at most 2×10^{-26} of its largest term at 25 digits at $t=1/2$ and $13/10$, while a 1% change in s_0 gives 10^{-3}. Both annihilations are derived, not only measured. With $W=R_3R_1(\theta+2)$, the Mellin operator is N_2W, and $N_2WG=0$ holds with no source term. Each term $t^{2i}Q_i(\theta)G$ of that operator is an integral on the line $\operatorname{Re}s=c-2i$, $2<c<4$; moving it to c crosses the poles of the Mellin transform of G at $s=4,6,8$, which come from the series of J^2 at $x=0$. The six residues cancel exactly, though none of them vanishes alone. The recurrence itself has no boundary term, since the relation for J^2 kills constants. The same bookkeeping gives the nonzero sources $-16/t$ for $\int_0^\infty xK_1(tx)J(x)\,dx$ and $-9216\,t^2$ for $\int_0^\infty xK_0(tx)J(x)^2\,dx$, both equal to quadrature at 25 digits. So $y=WG$ solves $N_2y=0$. In the normal form $v''+qv=0$ of N_2, $q=6-59/(4t^2)+O(t^{-4})$, with $q\ge5$ for $t\ge3.42$ and $q'=O(t^{-3})$. So by Sturm comparison every nonzero real solution has infinitely many zeros there, and its amplitude does not fall faster than $t^{-1/2}$. A function with an expansion in powers and logarithms of t at infinity has neither property, and y has one; so $WG=0$. For the chain the residues at $s=0,2,4$ cancel the same way, and $q\to20/9$ with $q\ge2$ for $t\ge5.32$, so $WC=0$ for its $W=R_3R_1(\theta-2)$. Since $R_1S=1+NR_3$ with N rational, $R_1P=-N\,R_3R_1(\theta+2)$, so P sends

every solution of $R_3R_1(\theta+2)$ to a multiple of $t/(1-t^2)^4$, which is odd in t. Near $x=0$, J is a sum of odd powers of x times powers of $\log x$. So to the right of its strip the Mellin transform of G has poles only at $s=4,6,8,\dots$, and the expansion of G at large t has only even powers of t, times logarithms. The coefficients of P are rational in t^2, so PG has the same property. On $t>1$, where R_1 is regular, PG is a single multiple of $t/(1-t^2)^4$, so that multiple is zero. Since G is analytic on $\operatorname{Re} t>0$ and P has rational coefficients, PG is meromorphic there; it vanishes on $t>1$, so $PG=0$ for all $t>0$. Equivalently, among the solutions of $R_3R_1(\theta+2)$, $\ker P$ is the set that are even in t at infinity, and at infinity P keeps exactly the integer exponents of $R_3R_1(\theta+2)$, in w. The analyticity of G at $t=1$ alone, or at $t=2$ alone, does not give $PG=0$: P has a solution with a logarithm at $t=1$ and one with a half-integer exponent at $t=2$. For the chain, $C=(8t^3/\pi^3)\int_0^\infty \chi(x)^2K_0(tx)\,dx$, where $\chi=\int_0^x sK_1$ is, near $x=0$, a sum of odd powers of x times powers of $\log x$. So $\int_0^\infty \chi^2K_0(tx)\,dx$ has only odd powers of t at infinity, times logarithms, and with the factor t^3 the function C has only even ones; the same steps then run with the solution $t^3/(1-t^2)^2$ of its R_1. The operator $R_3R_1(\theta+2)$ is singular at $t=1$, the equal-scale point where the fibre collapses (Sec. III B); its exponent -3 there comes from R_1, so the singularity is not apparent. Caps: the no-split search is modulo a prime, and its bound rests on the leading-term argument rather than the written proof the chain has; the annihilation and the parity step use the standard Mellin contour shift. Records: `docs/predictions/2026-10-19_p3e2_widened.md`, `docs/predictions/2026-10-20_slice_artanh.md`, `docs/predictions/2026-10-20_slice_parity_proof.md` and `docs/predictions/2026-10-21_wg_closure.md`. Guard: `tests/test_paper63_chain_slater_slice.py`.

[SYMBOLIC + MEASURED] At $\gamma=1$ the correlated triangle does not drop. Put the exponential geminal $f=e^{-\gamma r}$ on the two edges at one density, $Y=\langle \rho_i\rho_j\rho_k\, f_{ik}f_{jk}/r_{ij}\rangle$, at the fibre with scales $(1,1,t)$ and t on the density that carries both geminals. The registered prediction was that Y stays on the elliptic rung; it fails. The loop is still elementary: it is a mass derivative of the loop with masses $(\gamma,\gamma,0)$, a solid angle, $\pi^2\arctan(\sqrt{\det\mathcal{G}}/\mathcal{D})/\sqrt{\det\mathcal{G}}$ in the Gram data $\mathcal{G}$, $\mathcal{D}$ of the masses and momenta, matched by direct integration at five triples to 3×10^{-14}. But it has no uniqueness step. The two densities and the two geminals around it form a kite, whose Taylor series satisfies an order-two operator, measured on 600 coefficients and mapped exactly to the Legendre equation: the kite is elliptic. At $\gamma=1$ the minimal annihilator of Y has order four and divides exactly as $\tilde R_3\tilde R_1$, with factors of orders three and one that are not those of the chain. The piece $\tilde R_1$ has an algebraic solution. The piece $\tilde R_3$ is the adjoint of an operator whose solutions are the image of the solutions of F_3's operator under an exact order-two map, and since F_3's operator is anti-self-adjoint, $\tilde R_3$ is equivalent to it. Applied to Y, the order-four operator leaves at most 2.3×10^{-26} of its largest term at four values of t, with no source term, and no operator of order three annihilates Y: an exact search within a bound set by the local exponents finds no split. Attaching the last density turns the elliptic kite into the K3 family, a Clausen-type step. So at $\gamma=1$ the correlated triangle sits with the three-density chain, on the K3 rung, and not with the Coulomb triangle. At $\gamma=3/2$ an annihilator of Y of order nine has an exact composition series whose pieces have orders one (four of them, each with an algebraic solution), two and three. The order-three piece has the local data of F_3's operator twisted by t, with genuine singular points only at $t=0,\pm2,\infty$, which fixes it up to equivalence by Levelt's rigidity, in the form of Beukers and Heckman [35]; an exact order-two map from F_3's operator confirms the equivalence. The order-six right factor below the order-three piece leaves between 1.3 and 2.4 times the largest term of Y at three values of t, so Y needs the K3 piece, and no piece above K3 occurs (record `docs/predictions/2026-10-22_correlated_triangle_g32.md`). The order-two piece is elliptic. In $x=t^2$ it is equivalent, by an exact map of order one, to the operator of a period of the rational elliptic surface $Y^2=X^3-3x^2(16x^2+152x+25)\,X-2x^3(4x-5)(16x^2-616x+25)$, whose j-invariant is $(16x^2+152x+25)^3/\big(x(4x-1)^2(4x-25)^2\big)$ and whose singular fibres are I_1^*, I_2, I_2, I_1 at $x=0,1/4,25/4,\infty$; over t they are six fibres I_2, at $t=0,\pm1/2,\pm5/2,\infty$, that is at $t=|1-\gamma|$ and $1+\gamma$. The monodromy of the piece has integer traces, measured at 80 digits, and the map is exact (record `docs/predictions/2026-10-22_c2b_identification.md`).

[SYMBOLIC + MEASURED] The curve is Legendre's, with modulus $\lambda=(\gamma^2-(t-1)^2)/(4t)$, so that $\lambda(1-\lambda)$ is $-1/(16t^2)$ times the Källén function of t^2, 1 and γ^2. Twisted by t, the Legendre operator of degree $-1/2$ in $z=(t^2+1-\gamma^2)/(2t)$ is the period operator above, coefficient for coefficient. **[PRIOR ART]** The function itself is $\pi\bar S_3(a,b,w)$ of Ref. [22] (their Eq. 30), given there as a convolution of two square-root propagators, $(\pi/4)\int_{-\infty}^{\infty}dk/\sqrt{(k^2+a^2)((k-c)^2+b^2)}$, and as a complete elliptic integral of modulus $\sqrt{(c^2+(a-b)^2)/(c^2+(a+b)^2)}$; the Legendre form, the imaginary momentum and the triangle are the readings added here. It annihilates $C(t)=\int_0^\infty K_0(tx)K_0(x)\cosh(\gamma x)\,dx$, which converges for $t>\gamma-1$, the pseudo-threshold, to 5×10^{-41} at 40 digits, and $C=\pi^2P_{-1/2}(z)/(4\sqrt{t})$ to 2×10^{-16} at 15 digits: the continuation to imaginary momentum of a classical cosine transform of K_0K_0. Since $\int dp_2\,e^{ip_2x_2}=2\pi\delta(x_2)$, C is π times the two-dimensional bubble with masses t and 1, integrated over a line of external momentum moved off the origin by $i\gamma$; at $\gamma=0$ it is F_2. In the dictionary of Sec. III C the geminal therefore acts as an imaginary external momentum, and the singular points are where $\gamma=t+1$ or $\gamma=|t-1|$, the bubble's threshold and pseudo-threshold. The registered candidate with $e^{-\gamma x}$ in place of $\cosh(\gamma x)$

is not a solution, because its sinh part is not (record `docs/predictions/2026-10-22_c2b_feynman.md`).
This fixes the operator class of the piece, not the component of Y along it. **[SYMBOLIC + MEASURED]** At $\gamma = 2$ the reading was registered before Y was computed there, and it held piece by piece. An annihilator of order nine, fitted from 800 series coefficients and checked on Y to 5×10^{-26}, has the same composition series as at $\gamma = 3/2$. Its order-two piece is equivalent by an exact map to the Legendre operator with $z = (t^2-3)/(2t)$, its order-three piece is the twisted K3 piece at $t = \pm 2$ again, and its order-one pieces are algebraic, branched at $t = 2$ and $t = 2\gamma = 4$ (record `docs/predictions/2026-10-22_c2b_feynman_g2.md`; guard `tests/test_paper59_c2b_feynman_g2.py` for the order-two piece; the algebraic pieces, the order-nine annihilator and the K3 map on the record only). **[SYMBOLIC + MEASURED]** Why the order drops at $\gamma = 1$. In Feynman parameters the kite's density pair enters through $M(u) = \sqrt{1 + u(1-u)z}$ and its geminal pair through $N(v) = \sqrt{\gamma^2 + v(1-v)z}$. At $\gamma = 1$ the two are the same function: the two bubbles have equal masses, and the four pair thresholds $t = 2$, 2γ, $1 + \gamma$, $|1 - \gamma|$ fall onto 2 and 0. Two readings were registered and separated before computing: equal masses, or merely colliding thresholds. At $\gamma = 3$ the pseudo-threshold $|1 - \gamma|$ falls on the density-pair threshold $t = 2$ with unequal masses. There the annihilator of Y keeps order nine and the series $1, 1, 1, 2, 3$, the elliptic piece is present, mapped exactly from the Legendre operator with $z = (t^2 - 8)/(2t)$, and the genuine singular points are exactly $t = 2, 4, 6$; at $t = 2$ the K3 piece's half-integer exponents and the elliptic piece's I_2 fibre sit together, and no factor is lost. At $\gamma = 1/2$, before the branch point crosses the contour, the order is nine as well, with the Legendre map at $z = (t^2 + 3/4)/(2t)$ and singular points at $t = 1/2, 1, 3/2, 2$. So colliding thresholds remove nothing, and the drop at $\gamma = 1$ is the equal-mass degeneration, the step that makes the equal-mass sunrise special in the Feynman-integral literature [25]. The kite's operator in z has order six at $\gamma = 1/2$, $3/2$, 2 and 3, with genuine singular points exactly at minus the squared pair masses and at 0 and ∞, and order two at $\gamma = 1$. The annihilators are checked on Y to 6×10^{-27} at $\gamma = 1/2$ and 3×10^{-24} at $\gamma = 3$, and the maps are exact (record `docs/predictions/2026-10-23_gamma1_mechanism.md`; guard `tests/test_paper63_mg_mechanism.py` for the order-two pieces, the order-nine annihilators and the K3 maps on the record only). Cap: the mechanism is read from the integrand and tested by its consequences at two values of γ; no symbolic derivation of the drop from $N = M$ is given. **[SYMBOLIC + MEASURED]** The cusp-correct geminal $f = re^{-\gamma r}$ of the LiH engine, on both edges at $\gamma = 3/2$ and 2, brings no new type, as registered before computing from the fact that $re^{-\gamma r}$ is a γ-derivative of the same Yukawa triangle. Its annihilator of order nine, checked on the integral to 10^{-28}, has the same composition series: four algebraic pieces, branched at $t = 2$ and 3; an order-two piece, a different operator from the exponential case's, equivalent by an exact map to the same Legendre family; and the twisted K3 piece (record `docs/predictions/2026-10-22_geminal_shapes.md`; guard `tests/test_paper63_ga1_geminal.py` for the order-two piece and its difference from the exponential case's; the algebraic pieces, the order-nine annihilator and the K3 map on the record only). At $\gamma = 2$, registered as a check of the rule, the order is nine again with the same series and the same degrees, the genuine singular points are exactly $t = 1, 2, 3, 4$, the order-two piece is again a different operator from the exponential case's and maps exactly from the Legendre operator with $z = (t^2 - 3)/(2t)$, and the algebraic pieces carry the exponents $-11/2$ and $-13/2$ at $t = 2\gamma$ as at $\gamma = 3/2$; the operator is exact, from coefficients modulo twelve primes and rational reconstruction, and leaves 2×10^{-24} on the integral at 32 digits (record `docs/predictions/2026-10-23_geminal_shapes_g2.md`; guard `tests/test_paper63_ga3_geminal.py` for the order-two piece and the algebraic pieces, the order-nine annihilator and the K3 map on the record only). **[SYMBOLIC + MEASURED]** One function carries both elliptic pieces met so far. Write $\mathcal{L}(a, b; c) = \int_0^\infty K_0(ax) K_0(bx) \cos(cx)\, dx = \pi^2 P_{-1/2}(z)/(4\sqrt{ab})$, with $z = (a^2 + b^2 + c^2)/(2ab)$. At $c = 0$ it is the radial core of the three-center integral at zero separation: with $a_i = c_i^{-1/2}$ it reproduces Eq. (2) to 4×10^{-31} at three pairs of scales (two in the guard, to 10^{-20}). At $c = i\gamma$ it is $C(t)$ above. Zero or real momentum gives $z \geq 1$. Imaginary momentum gives $z = \cos\theta$ between the pseudo-threshold and the threshold, $|t - 1| < \gamma < t + 1$, with θ the angle opposite the side γ in the triangle with sides t, 1 and γ, and the modulus is $\lambda = \sin^2(\theta/2)$. The thresholds are the flat triangles, since the Källén function is -16 times the squared area. The slot is not a convention: with the geminal as a mass and a density scale as the momentum, the two other assignments give the families $j = 4(16x^2 + 472x + 25)^3 / \big(9x(4x-1)^2(4x-25)^2\big)$ and $j = 4(16x^2 - 104x + 601)^3 / \big(9(4x-1)^2(4x-25)^2\big)$, and neither is isogenous to the piece's, since their traces of Frobenius differ at most good primes at every rational x tried, while the two solutions of the degree-six search agree at every one. The right angle, $\gamma^2 = t^2 + 1$, gives $\lambda = 1/2$ and $j = 1728$, the same complex-multiplication value $K(1/\sqrt{2})$ as the three-center fibre at $\rho = 1/2$. **[OBSERVATION]** On this family, then, composing unequal decay rates costs one elliptic period: at zero momentum its modulus is set by their ratio, and equal rates make it elementary, as Eq. (2) records; at imaginary momentum it is set by the angle of their triangle, and a flat triangle is singular. Reading one-loop integrals through triangles of masses and momenta is the approach of Davydychev and Delbourgo [37]; what is added is its appearance in a correlated molecular integral. **[SYMBOLIC + MEASURED]** The shape

of the geminal matters at this level. A Gaussian geminal's avatar, $\int_0^\infty K_0(tx)K_0(x)e^{-x^2}dx$, is annihilated by $t^4(\theta+1)^2 - t^2\theta^2(4\theta+1) + 4\theta^2(\theta-2)^2$ to 10^{-41}, and its monodromy companion, with I_0 for $K_0(tx)$, grows like $e^{(t-1)^2/4}$, so every annihilator has an irregular singular point at infinity: at the level of the one function, Slater geminals stay on the ladder of Picard–Fuchs operators and Gaussian ones leave it for exponential periods (registered before computing, record `docs/predictions/2026-10-22_geminal_shapes.md`).
[**SYMBOLIC + MEASURED**] On the full triangle, with e^{-r^2} on both edges and scales $(1, 1, t)$, the Gaussian shape removes more than the elliptic piece. The minimal annihilator of Y has order nine and x-degree 13 and splits exactly into two pieces, of orders four and five, both irregular at infinity, with formal solutions growing like $e^{t^2/4}$ and $e^{t^2/8}$; its only genuine finite singular point is $t = 2$. No piece is equivalent to F_3's operator, although each keeps the K3 piece's Jordan block of size three at $t = 0$: the K3 rung of the Slater triangle is not set by the densities and the Coulomb line alone. The prediction that the K3 piece survives failed. The operator leaves 2×10^{-27} on Y at 32 digits, and an independent evaluation of Y agrees to 4×10^{-14} (record `docs/predictions/2026-10-23_gaussian_geminal_triangle.md`; guard `tests/test_paper63_gy_gaussian.py`). Caps: one value, $\gamma = 1$, the equal-mass value, at which the Slater triangle has no elliptic piece either, so the comparison of shapes is cleanest for the K3 piece; the irreducibility of the two pieces and the absence of an F_3 piece rest on bounded searches; and minimality assumes that $\sqrt{\pi/2}\, e^{1/2}\, \mathrm{erfc}(1/\sqrt{2})$, the constant that enters the Taylor coefficients, is irrational. Guards: `tests/test_paper63_one_formula.py`, `tests/test_paper63_geminal_shapes.py`. The singular points of Y's annihilator move with γ, to $t = |1-\gamma|, 2, 1+\gamma, 2\gamma$, three of which meet at $t = 2$ for $\gamma = 1$. [**OBSERVATION**] The drop of the Coulomb ring is therefore read as a property of uniqueness, not of rings: both loops integrate out exactly, but only the Coulomb one factors into pairwise inverse distances, and only it lowers the rung.

[**OBSERVATION**] The registered decision rule did not anticipate this outcome. It allowed the Coulomb triangle to land on the chain's rung, with or without an elementary term, or to climb; it lands below. By the rule's letter L_G is an order-three factor that is not equivalent to the chain's operator, which the rule called a climb, but L_G factors into lower pieces, so it is not a climb in the sense the record intended. Two side readings failed. The singular points at $t = \pm 1$ were not predicted; they belong to the elliptic piece. And a Coulomb ring of three does not sit with the chain of three, as the record expected, but lower. The count of linked densities orders the trees, not the rings. The caps of Sec. III C apply: these are fibre statements, and the integrated value may sit lower. Further caps: for the correlated triangle, the exponential shape at five values of γ, the first the equal-mass value, with t on the density carrying both geminals, and the LiH engine's cusp-correct shape at $\gamma = 3/2$ and 2 only, and a Gaussian shape at $\gamma = 1$ only; the kite operators were guessed from series and checked on their coefficients, and are now also derived by integration by parts (below), and the mechanism of the drop at $\gamma = 1$ is tested by its consequences at two values of γ, not derived; only the triangle among rings; and that the chain satisfies its own order-four P rests on the Mellin contour shift. The search that excludes a K3 piece from the triangle is bounded in degree, unlike the chain's no-split proof, but the orders settle it: the factors found have orders one and two, which leaves no room for the order-three K3 factor. Guards: `tests/test_paper63_density_ring.py`.

[**PRIOR ART**] The one-center values are Remiddi's [7]: the three-electron correlation integral over Slater functions closes in logarithms and dilogarithms, and Fromm and Hill [6] treat the general case. The uniqueness relation (7) is classical; its general-dimension form, which gives the constant π^3 at $D = 3$, and the early references are collected by Preti [39]. Under the relation of Sec. III C to the banana ladder, the triangle's elliptic piece is the $n = 2$ rung itself. What is added here is the fibre reading: the Struve moment (10), its operator, its place below the chain, and the correlated triangle's place with it. The solid-angle form of the massive loop is Davydychev and Delbourgo's (Sec. 5.1 of [37], which recovers Nickel's result [38]); here it also rests on its numerical check. No literature search was made for the operator (11).

[**SYMBOLIC + MEASURED**] The kite operators above were guessed from Taylor series. They also follow from integration by parts, the standard reduction of Feynman integrals [8, 9]. In momentum space the kite is a two-loop integral in $D = 3$ with five lines: the two densities with index 3/2, the two geminals $(q^2+\gamma^2)^{-2}$ and the Coulomb line q^{-2}. A reduction modulo a large prime, sampled at many values of z, gives the differential system of the master integrals, and the kite's cyclic vector in that system gives its operator. At $\gamma = 1/2$, 3/2, 2 and 3 the operator has order six and equals the guessed one exactly. At $\gamma = 1$ the master integrals drop from nine to three and the operator is the Legendre one, so the equal-mass degeneration shows in the reduction itself. On the tetrahedron, the three-loop vacuum integral with the apex density of mass t, the reduction gives an operator for Y of order ten at $\gamma = 3/2$, divisible on the right by the order-nine operator above. Off the symmetric slice, at density scales $(1, 2)$ and geminal exponents $(3/2, 3)$, the series route found no operator with 260 unknowns or fewer. The reduction gives one of order eleven and degree 113 in z, about 1300 unknowns. Its integer coefficients reach 565 bits, but the eighteen-dimensional system behind it needs only 37, so the system is rebuilt exactly from two primes and checked on a third, and the operator is built from it over $\mathbb{Q}(z)$. It agrees with

the operator computed at thirteen primes, and on 300 Taylor coefficients at 280 digits it leaves 1.3×10^{-280} of the largest term; with one coefficient changed by $1/200$ it leaves 4×10^{-41}. Its genuine singular points are exactly $z = -1/4, -1, -9/4, -4, -9, -49/4, -16, -81/4$, with 0 and ∞: minus the squares of the two-line cuts through the two densities and through the two geminals, and of the three-line cuts that pair each density with the geminal on the opposite edge. The points $z = -25/4$ and -25, where a pairing of each density with its adjacent geminal would put its cuts, are ordinary, and the remaining 92 roots of the leading coefficient are apparent. Controls: the one-loop bubble in $D = 3$ against its arctan closed form, and the kite with the Coulomb line removed, whose exact Taylor series is the product of two bubbles (records `docs/predictions/2026-10-24_ibp_kite.md` and `docs/predictions/2026-10-23_price_formula_slots.md`). **[SYMBOLIC + MEASURED]** On the tetrahedron at the same point the reduction needs 34 masters and gives an operator for Y/t^3 in $x = t^2$ of order nineteen, exact over $\mathbb{Q}$ (the system rebuilt from two primes and checked on a third). On Y from a 60-digit grid it leaves 5×10^{-55} to 3×10^{-43} of the largest term, following the quadrature error. The genuine singular points of Y's minimal operator in $x > 0$ are the same eight cuts, $x = 1/4, 1, 9/4, 4, 9, 49/4, 16, 81/4$. The other roots of the leading coefficient are apparent, and at each of the eight some analytic continuation of Y changes on a small loop. The change is the same to four digits in 64- and 90-digit arithmetic, and at least 10^{10} times the change on a loop around an ordinary point along a path at least as long. **[SYMBOLIC + MEASURED]** The order nineteen is not minimal. The reduction treats the two sectors with one geminal and no Coulomb line as different, but each is the same sunrise of the three densities times a one-line tadpole, so their masters differ by the constant ratio of the geminal scales, $\gamma_1/\gamma_2 = 1/2$. With that relation imposed the operator has order fifteen, exact over $\mathbb{Q}$, and on Y it leaves 2×10^{-56} to 4×10^{-46} of the largest term; the same construction with the ratio $1/3$ leaves 10^{-4} to 2×10^{-2}. The reduction's system is block triangular in its sectors, so the pieces can be read sector by sector, and they are the predicted ones. The K3 operator of the chain of three with scales $(t; 1, 2)$ is the sunrise sector's own block, and it is a left factor. The Legendre family pulled back to the two crossed pairings, $z_1 = (4t^2 + 7)/(16t)$ and $z_2 = (t^2 - 8)/(2t)$, gives the two pieces of order two, from the sectors with one geminal and the Coulomb line; the direct pairings do not occur. The other seven pieces have order one and algebraic solutions, branched at $x = 1, 9$, at $9/4$ and at $81/4$, and come from the sectors with both geminals. Each of the three identifications is an exact map over $\mathbb{Q}(x)$, rebuilt from many primes and checked by exact division. Y needs the K3 piece: the right factor without it leaves 2×10^{-4} to 4×10^{-3} of the largest term. It needs both elliptic pieces, since its minimal operator is genuinely singular at $x = 1/4$ and $x = 4$ and no other piece is singular there. **[SYMBOLIC + MEASURED]** The minimal operator of Y has order thirteen, exact over $\mathbb{Q}$. Two of the three order-one pieces branched at $x = 9/4$ are left factors of the order-fifteen operator. The pairing of Y with each of them, the bilinear concomitant, is a constant, and on Y it is below 10^{-56} of its largest term and grows with the quadrature error; the same pairing with the K3 piece, the control, is 10^{-8} of its largest term and does not move at twice the step. So the right factor of order thirteen without those two pieces annihilates Y, leaving 8×10^{-57} to 2×10^{-46} of the largest term. Its only irreducible left factor is the K3 piece, once, and Y needs that piece, so no smaller operator annihilates Y. Its five order-one pieces are branched at $x = 1, 9$ (one), at $9/4$ (one) and at $81/4$ (three). The K3 piece is irreducible: its exponents leave no room for a factor of order one or three, and its exterior square has no element of the one admissible form, so it has no factor of order two. Its monodromy preserves a symmetric form and no alternating one, with reflections at $x = 1$ and $x = 9$, and its Jordan blocks are $2 + 2$ at $x = 0$ and $3 + 1$ at infinity. Caps: the reduction is new code, checked against the guessed operators and the series; the two vanishing pairings are measured, not proved; the left-factor searches behind the order thirteen are bounded, with planted controls found; and the form, the blocks and the irreducibility are computed modulo two primes, the last by searches that are complete for its exponents. An earlier reading of the order-nineteen operator, as one piece of order twelve with none of these pieces at either end, came from a map search whose box was too small at the left end, and is withdrawn (records `docs/predictions/2026-10-23_price_formula_slots.md`, addendum 6, `docs/predictions/2026-10-25_geometric_route_y.m` and `docs/predictions/2026-10-25_tetrahedron_route_y.md`). **[PRIOR ART]** Integration by parts on this tetrahedron in momentum space is Puchalski, Kędziera and Pachucki's, for the three-electron Slater integral with integer indices [12], which they reduce to a differential equation. What is added here is the family with half-integer density lines and the geminal's squared lines, and the check against operators found independently. Guards: `tests/test_paper63_ib_kite.py` and `tests/test_paper63_y_minimal_order.py`.

[MEASURED] The operators are also evaluators. The kite's operator, with the kite's first six Taylor coefficients at $z = 0$ as its only input, gives the kite at all 247 nodes of the grid behind Y. It does so by Taylor steps along a path that passes above the three apparent singular points on the positive axis. Each step reaches half the distance to the nearest root of the leading coefficient, apparent roots included, because rounding excites solutions of the recurrence that blow up there. On one process this takes 64 s, of which the input takes 1 s; the quadrature it replaces took 7894 s on fifteen. Y from the two routes agrees to 2×10^{-69} at four values of t, the precision at which the grid

is stored. The node error grows towards large w, to 3×10^{-8} near $w = 10^{12}$, but those nodes carry so little weight that no node moves Y by more than 1.4×10^{-68}. With the derivation included, the route costs about 9300 s of processor time, about a thirteenth of the upper bound on the quadrature's. The derivation is three reduction runs on six worker processes, one of which was measured at 3065 s of processor time, and a 13 s lift. Changing one operator coefficient by 1/200 moves Y by 1 to 7%. Y's own operator does not pay in the same way. From the grid's data at $t = 1.7$ it reaches five other values of t to 4×10^{-48} or better, but at 4 to 12 s per value, against 0.02 s for one more grid sum, because once the kite is known the outer integral is a cheap sum. At 60 digits the structure pays where the cost is, in the inner kite. Caps: one parameter set, so a new configuration needs a new derivation, which dominates the cost; times on one machine, with the quadrature's processor time an upper bound; the comparison is to this paper's own grid, not to a production Slater-integral code such as Ref. [10]; and no certified error bounds, only agreement with the grid (record `docs/predictions/2026-10-26_operator_evaluator.md`, registered before computing; guard `tests/test_paper63_operator_evaluator.py`).
[**MEASURED**] At chemistry precision the comparison reverses. For Y to 10^{-12}, a double-precision tanh-sinh quadrature of the same two-dimensional integral, graded at the ends and split along its ridge, takes 0.033 s of processor time on one core. The operator route takes 16 s with the operator already given, about 500 times slower, and 3×10^5 times slower with its derivation counted. It also cannot run below 60 digits: at 20 to 40 digits its Taylor walk diverges. The two routes agree on Y to 7×10^{-14}. So the operator is an evaluator only where high precision is wanted; for chemistry it is a statement about structure, not a shortcut (record `docs/predictions/2026-10-28_kite_speed_test.md`, registered before timing; one parameter set, one machine). [**PRIOR ART**] Evaluating Feynman integrals by solving their differential equations in series is standard in particle physics [13–15], and Ref. [12] reduces the integer-index three-electron integral to such an equation. What is added is the measured cost on this family.

[**MEASURED**] Most of such an operator's singular points can be read off before it is derived, but not all of them. At a second set of rates, chosen blind, $(a, b, \gamma_1, \gamma_2) = (1, 3/2, 7/3, 8/3)$, the kite's operator again has eighteen masters and order eleven. The eight singular points predicted from its cuts, at minus the squared sums and differences of the rates on each cut, are all genuine and carry the predicted local exponents; the pairings of adjacent lines give none. But a ninth genuine point appears at $z = -111/1120$, with exponents $-7/2, 0, 1, \ldots, 9$, and the leading Landau equations of the kite and of its contractions have no solution there. A count of critical points of the Lee–Pomeransky likelihood, which equals the number of masters at a generic point, confirms it: 22 at generic z, 21 there, at two primes. The drop is in the full kite only. Its position has a closed form in the squared rates $A = a^2$, $B = b^2$, $G_1 = \gamma_1^2$, $G_2 = \gamma_2^2$:

$$z_9 = \frac{(AG_2 - BG_1)(A - B - G_1 + G_2)}{(A - G_1)(B - G_2)}. \qquad (14)$$

It is symmetric under the kite's reflection $(a, \gamma_1) \leftrightarrow (b, \gamma_2)$ and under $(a, b) \leftrightarrow (\gamma_1, \gamma_2)$. Its poles lie where an adjacent pair has equal rates, the pairings the cuts ignore. At the original rates $a\gamma_2 = b\gamma_1$, so $z_9 = 0$: the ninth point sits on $z = 0$ there, which accounts for that operator's exponent at $z = 0$ that the cuts did not explain. The formula was fitted from 41 sets of rates, 29 unknowns, and passed 18 held-out sets at two primes; it reproduces $-111/1120$ at the blind point, and on the family $(1, 2, 3/2, \gamma_2)$ the critical-point count drops from 22 to 21 exactly at z_9 (record `docs/predictions/2026-10-26_kite_ninth_point.md`, registered before computing). [**PRIOR ART**] The location is known. Broadhurst gives it, with $s = -z$, as the point s_3 at which the two-particle cuts of the kite are evaluated when its rung becomes massless with unequal adjacent masses (Ref. [11], Eqs. (24)–(25)), without naming it. The triangles' leading Landau singularities $s_\pm$ are also his (Eq. (9) there); what is ours is the identification of s_3 as the massless-rung limit of one of them, for the triangle formed by the two lines of one cut and the rung, the other two lines on shell, which the kite's own leading Landau equations do not contain.
[**MEASURED**] A registered test confirms that this is the mechanism here. With all five lines on shell and the loop momenta in one plane, eliminating the equations in components gives exactly the factor of Eq. (14) at three sets of rates and no other. The multipliers are all nonzero, and the first Symanzik polynomial U vanishes on them, which is why a Landau test that requires $U \neq 0$ misses the point. Given a small Coulomb mass, the point does not vanish but moves: the count drops on a curve through $(z_9, 0)$, one of Broadhurst's two triangle singularities $s_\pm$, while the other runs to infinity as the mass goes to zero. A scan in z that assumes no location finds, at the blind rates, exactly ten drop points: $z = 0$, z_9 and the eight cut values. Caps: the massive curve is checked modulo two primes at three masses, not over the rationals, and the local shape, the drop size on the curve and the escape direction were not run (record `docs/predictions/2026-10-27_kite_ninth_point_origin.md`, registered before computing; guard `tests/test_paper63_kite_ninth_point.py`, which also covers the blind point and Eq. (14)). What is added here is the measurement in this family: the critical-point drop, the exponent, and the complete drop locus at one set of rates. The operator's degree, 120, exceeds the prediction, and its coefficients needed a fourth prime. With γ_2^2 left free the differential system has degree 16 in it, but the kite written in the masters has degree 26, so the one-parameter family exceeded

its registered sample cap and was not built (record docs/predictions/2026-10-26_parametric_kite_operator.md, registered before computing). [**PRIOR ART**] Reading singular points from cuts is standard Landau and threshold analysis.

[**MEASURED**] In the molecular calculation the triangle class is a modest share of the cost. In the explicitly correlated LiH calculation of Paper 19 at its best truncation, the class that Y models takes 21% of a one-process run, third of its four large pieces, behind the electron–nucleus and kinetic blocks. The part an operator evaluator would replace, the factorization of the three-body kernels, is 3% of the run; the rest of the class is about 2.4 million Coulomb mode solves. So a free triangle class would speed the run by at most 1.27 times, and a free kernel build by 1.03 times. Caps: one run per truncation on one machine; a reduced truncation is no proxy, giving 48% (record docs/predictions/2026-10-26_lih_triangle_share.md, registered before computing).

[**MEASURED**] Whether the one-mass slice could be tabulated, as Gaussian codes tabulate the Boys function, depends on how many master integrals it needs and in how many variables. Counting critical points, as for the kite, the slice needs four masters, and the index range the production code uses needs a fifth, an antiderivative in D from a pole at $x = 0$; a scaling identity leaves two variables. By the registered gate this is borderline: a table is not ruled out, but its speed against a production code [10] is untested. The physical fibre, with two offsets, needs six masters in three variables. Caps: the scaling identity missed its registered precision, 2.7×10^{-25} against 10^{-25}, and an unregistered rerun agrees to 1.6×10^{-37}; these are counts of masters, not the cost of a table (record docs/predictions/2026-10-27_master_count_tabulation.md, registered before computing; guard tests/test_paper63_master_count.py).

[**MEASURED**] Production explicitly correlated codes do not use a single Gaussian geminal. For many integrals they replace $e^{-\gamma r}$ by a least-squares fit, six Gaussians by default. The off-slice triangle above, known to about fifty digits, measures that replacement. With the fit in the plain norm $\int_0^\infty r^2(\cdot)^2\,dr$, which at three and six terms gives the STO-3G and STO-6G exponents [16], the six-term fit makes Y too small by a relative 3.6×10^{-5} at $t = 1/2$ down to 2.4×10^{-6} at $t = 4$. The error falls like $e^{-b\sqrt{n}}$ in the number of terms n, the law the cusp at $r = 0$ sets for Gaussian expansions of exponentials [17], proved as an upper bound in a weighted norm by Braess [18]; the constant b on this integral is our measurement. With an extra weight $e^{-\gamma^2 r^2}$ the fit loses its tail, and the six-term error is 5.4×10^{-2} to 1.7×10^{-2}. So on this integral the error is set by how far out the fit is trusted, not by the cusp. Caps: the production weight function is not reproduced, so these two weights do not price a production fit; one configuration, with model densities (record docs/predictions/2026-10-25_gaussian_fit_geminal.md, registered before computing; its prediction that the weighted fit stays within 10^{-3} missed). [**MEASURED**] A fit that a production code ships, the six-term fit written into Psi4's source [20], is the best of the three at the cusp and the worst in the tail. It is 1.8% low at $r = 0$, against 6% for the plain fit, and half the true value at $\gamma r = 4$. On the same Y it gives an error of -13% at $t = 1/2$ down to -5.5% at $t = 4$, more than three thousand times the plain fit's. The three fits are ordered by their tails, not by their cusps. Caps: Y weights the geminal over its whole range, while a correlation energy is dominated by short range, so this prices the fit on this integral and not on an F12 energy; the fit's weight and its literature source are not verified, nor which integrals the code uses it for (record docs/predictions/2026-10-26_production_geminal_fit.md, registered before computing; guard tests/test_paper63_production_geminal_fit.py).

IV. THE GRAPH READING, AS A BOUNDED CONJECTURE

[**OBSERVATION + PRIOR ART**] Read as a Feynman graph, the cases of Paper 59's geminal section and of Sec. III D fit one pattern. The densities and the kernels are the lines; a Slater geminal is a line whose mass is its exponent γ, and the Coulomb kernel is a line with no mass. On every instance tested the class is set by the shape of the graph together with the masses on its lines. A massless Coulomb line adds no threshold, so for Coulomb integrals the density topology alone sets the rung, as Paper 59 states. A Slater geminal keeps the curve the densities build, but it lifts Paper 59's same-pair master to genus one, and the correlated triangle's operator drops order where the geminal's mass equals that of a density pair, $\gamma = 1$. A Gaussian geminal is not a propagator of this kind, and its integrals leave the Picard–Fuchs ladder (Sec. III D). That the internal masses set the geometry of a Feynman integral is standard: the sunset graph with massless lines evaluates in Gamma functions, and with massive lines it is elliptic [25, 34]. What is recorded here is only that these molecular integrals follow the same dictionary. The nearest theorem in shape is Brown's: for massless primitive ϕ^4 graphs in four dimensions the set of periods is minor monotone, so a minor-closed class of graphs bounds the ring the periods lie in [24]. That theorem has no masses; here the masses enter too. It is a pattern across the tested instances, not a derivation. It would fail on a Coulomb-only integral whose class depends on anything but the density topology, or on a Slater-geminal integral whose class does not change at an equal-mass value of its exponent (guards tests/test_paper59_h2_geminal_master.py, tests/test_paper63_geminal_shapes.py, tests/test_paper63_gy_gaussian.py,

`tests/test_paper63_mg_mechanism.py`).

The pattern is conjectured for every exponential-type orbital; the dictionary of Sec. II places each in its graph's $1s$ module, and the conjecture is bounded.

Conjecture 1 (the density-graph ladder)**.** *Let Γ be a connected density graph over exponential-type orbitals with Coulomb or Slater-geminal kernels, and let its fibre be the integral with all centers coincident. Then: (i) if Γ is a tree of n two-center densities, the minimal annihilator of the fibre's one-scale family has, among its composition factors, the operator of F_n of Sec. III C—for $n \leq 5$ the Hadamard product of the Legendre operator with the equal-mass banana with $n-1$ propagators—and nothing above it; (ii) the rung names that operator, not a finite basis of periods: the fibre is in general an iterated integral of its top piece against elementary kernels, and no finite set of periods of the top piece and the rungs below it is expected to contain the value; (iii) for a Coulomb ring, each loop that integrates out exactly by uniqueness lowers the rung by one density, and a geminal loop does not; (iv) the integrated value over separations and Feynman parameters has class at most that of its fibre.*

What is proved, what is measured, what is open. For $1s$ orbitals part (i) is proved at the level of the holomorphic period for all n (Eq. (5)), with the operator identified as the Hadamard product for $n = 2$ to 5 (Sec. III C); for every other exponential orbital Sec. II places the integral in the same D-module, and for s-type densities on the one-scale slice Proposition 2 proves the holomorphic period sits at or below rung n, and on it for $n \leq 8$ when it is not identically zero; the real three-dimensional integral reduces onto the Bessel-moment family at $n = 3$, derived on the Slater slice under the Mellin contour shift and shown on $(t; 1, 2)$ by numerical annihilation and a search modulo a prime, and at $n = 4$ by numerical annihilation and bounded searches; for $n \geq 5$ the three-dimensional reduction is not shown. Part (ii) is the lesson of the three-electron chain (Sec. III B): at equal scales the fibre collapses onto K3 periods and π^3, but at four generic scale triples integer-relation searches at 340 and 400 digits in five bases built from the K3 and elliptic periods, their derivatives, products, dilogarithms, trilogarithms and Tate numbers find nothing while every planted relation is recovered (three of the twenty searches void), and on both one-parameter slices computed the fibre is a K3 function integrated once against an elementary kernel, proved at the fingerprint weights under the Mellin contour shift and shown elsewhere by a bounded search modulo a prime. That null is the conjecture's stated test case: a relation found in a principled widening of the basis would show the fibre sits in a finite period ring after all and part (ii) would be withdrawn; a relation that needs a new period type would move the chain off rung three. The record is `docs/predictions/2026-10-19_p3e2_widened.md`. Part (iii) rests on one ring, the triangle, in which the Coulomb loop drops and the geminal loop does not (Sec. III D); the ring of four was not computed. Part (iv) is consistent with every instance: the two-center exchange class builds the elliptic fibre and closes at genus zero, and the integrated three-center value leaves the two-center class as a function of the bond angle (Paper 59). The conjecture fails on a Coulomb-only integral whose class depends on anything but the density graph, on a tree whose real three-dimensional reduction brings a piece above F_n, on a general-orbital fibre with a factor above its graph's $1s$ rung, or on a Slater-geminal integral whose class does not change at an equal-mass value of its exponent. The nearest theorem in shape remains Brown's [24], for massless graphs; here the masses enter, and a proof would have to run through the Symanzik polynomials of Γ with the Yukawa masses on its lines, which we have not attempted.

V. PLACEMENT IN THE LITERATURE

[PRIOR ART] The operators of the ladder are known. The Hadamard construction and the tables that place F_4 and F_5 at entries #58 and #39 are Almkvist and Zudilin's [29, 33]; the equal-mass banana operators are Vanhove's [34] and their holomorphic periods are Broadhurst's random-walk moments [40]; that an l-loop banana carries a Calabi–Yau $(l-1)$-fold is Bönisch, Fischbach, Klemm, Nega and Safari's [43]; the two-scale and three-scale Bessel moments are Bailey, Borwein, Broadhurst and Glasser's [22], and the Kloosterman motives behind the Ising class are Fresán, Sabbah and Yu's [26]. Integration by parts on the three-electron tetrahedron is Puchalski, Kędziera and Pachucki's [12], and the location of the kite's ninth singular point is Broadhurst's [11]. What is new here is the route from molecular integrals to these objects, the tree identity that makes the holomorphic period of $1s$ densities depend on n alone, the dictionary that places every orbital in its graph's $1s$ module, the readings of the ring, the geminal and the Gaussian, and the prospective record.

[PRIOR ART] On the chemistry side the closest work is the Averys' momentum-space Coulomb–Sturmian program, whose monograph [3] and papers [5, 27, 28] treat exactly these integrals under the same Fock projection. Their route to a multicenter repulsion integral is to expand each two-center density in a one-center Coulomb–Sturmian basis of doubled exponent on its two centers, solving a linear system whose matrix is the overlap of that basis; the expansion is a series rather than an identity, its coefficient matrices are not sparse, and at short internuclear separation the system is ill-conditioned by the overcompleteness of the basis, which is handled by phasing out its small eigenvalues [27]. **[OBSERVATION]** In the language of this paper that route reduces every integral to the genus-zero one-center class by an infinite series, and the short-range difficulty it meets is where the series confronts the genus jump of Eq. (1): an elliptic period is not a finite combination of genus-

zero terms. Neither the Bessel-moment reduction nor the ladder appears in that program; this is an absence from a reading of the monograph's chapters on interelectron repulsion and of the cited papers, not a proof of absence. The overcompleteness itself is older than either program [4].

VI. SCOPE, AND WHAT IS OPEN

Four caps bound every statement above. First, the rungs are read on the fibre and its holomorphic period; the integrated value may sit lower, and in one class it does. Second, the reduction of the real three-dimensional integral to the Bessel-moment family is shown for the chain of three and the star and path of four; for five or more densities the holomorphic-period statement stands and the three-dimensional one is not shown. Third, the dictionary places a general-orbital integral in the D-module of its graph's $1s$ integral; it keeps the holomorphic period at or below the $1s$ rung only for trees of s-type densities on the one-scale slice, and on it there for $n \leq 8$ when the period is not identically zero; it bounds no fibre, and that every computed fibre sits on the $1s$ rung is observed, not proved. Fourth, among rings only the triangle was computed, and the generic three-electron fibre's content beyond the K3 and elliptic periods is evidence from integer-relation searches, not a theorem.

What is open is in two layers. Below the conjecture: the three-dimensional reduction for $n \geq 5$; the ring of four; whether a general-orbital fibre stays at or below its graph's $1s$ rung, which is open even for s-type trees, and whether it reaches that rung; and the principled widening of the basis of the generic-fibre searches. Above it: a proof through the Symanzik polynomials that the density graph with Yukawa masses has the motive the ladder measures. The finite closed form of the first rung's integrated value is Paper 59's open problem and Paper 61's subject, not this paper's.

VII. CONCLUSION

Every exponential-type orbital is a derivative of the $1s$ Yukawa function in its exponent and its center, so every molecular integral over such orbitals is a derivative of the $1s$ integral of its density graph and lies in that integral's D-module; for s-type trees on the one-scale slice its holomorphic period sits at or below that integral's rung, and on it for $n \leq 8$ when not identically zero; in every case computed the integral sits on it. On the graph, trees of n linked two-center densities share one holomorphic period, a Bessel moment whose one-scale operator, for $n = 2$ to 5, is the Legendre operator Hadamard the equal-mass banana with $n - 1$ propagators: elliptic at two densities, a K3 at three, a Calabi–Yau threefold at four, and nothing new at five. The integrals of a molecular Hamiltonian link two densities and, over s-type orbitals, stay elliptic at the fibre for every molecule; the class climbs only in correlated integrals that link more. A Coulomb ring of three does not climb and the correlated triangle does, read as uniqueness. A Slater geminal is an imaginary external momentum on this ladder and a Gaussian one leaves it. The operators are known; the route from chemistry to them, the tree identity, the dictionary and the prospective record are what this paper adds, and the conjecture that organizes them is stated with the case that tests it: the rung names an operator, not a number, and the generic three-electron fibre is where that must be decided.

[1] J. Loutey, "The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: A Genus-One Radial Core at the Third Center in Momentum-Space Slater Theory," GeoVac Paper 59 (2026).

[2] J. Loutey, "Forced and Free Content in the GeoVac Framework: An Observations Paper," GeoVac Paper 57 (2026).

[3] J. E. Avery and J. S. Avery, *Hyperspherical Harmonics and Their Physical Applications* (World Scientific, Singapore, 2018).

[4] H. J. Monkhorst and B. Jeziorski, "No linear dependence and multi-center integral problems in momentum space quantum chemistry," *J. Chem. Phys.* **71**, 5268 (1979).

[5] J. E. Avery and J. S. Avery, "4-Center STO interelectron repulsion integrals with Coulomb Sturmians," *Adv. Quantum Chem.* **76**, 133–146 (2018).

[6] D. M. Fromm and R. N. Hill, "Analytic evaluation of three-electron integrals," *Phys. Rev. A* **36**, 1013–1044 (1987).

[7] E. Remiddi, "Analytic value of the atomic three-electron correlation integral with Slater wave functions," *Phys. Rev. A* **44**, 5492 (1991).

[8] K. G. Chetyrkin and F. V. Tkachov, "Integration by parts: the algorithm to calculate β-functions in 4 loops," *Nucl. Phys. B* **192**, 159 (1981).

[9] S. Laporta, "High-precision calculation of multi-loop Feynman integrals by difference equations," *Int. J. Mod. Phys. A* **15**, 5087 (2000); arXiv:hep-ph/0102033.

[10] W. Chen, L. Silberman, U. Segal and C. Greif, "Fourier transform-based computation of two-electron four-center Coulomb repulsion integrals," arXiv:2610.12608 (2026).

[11] D. Broadhurst, "Multivariate elliptic kites and tetrahedral tadpoles," arXiv:2212.01962 (v2, 2024), to appear in the proceedings of LT-15 (Varna, 2023).

[12] M. Puchalski, D. Kędziera and K. Pachucki, "Ground state of Li and Be^+ using explicitly correlated functions," *Phys. Rev. A* **80**, 032521 (2009); arXiv:0909.5542.

[13] F. Moriello, "Generalised power series expansions for the elliptic planar families of Higgs + jet production at two loops," *JHEP* **01** (2020) 150; arXiv:1907.13234.

[14] M. Hidding, "DiffExp, a Mathematica package for computing Feynman integrals in terms of one-dimensional se-

ries expansions," *Comput. Phys. Commun.* **269**, 108125 (2021); arXiv:2006.05510.

[15] X. Liu and Y.-Q. Ma, "AMFlow: a Mathematica package for Feynman integrals computation via auxiliary mass flow," *Comput. Phys. Commun.* **283**, 108565 (2023); arXiv:2201.11669.

[16] W. J. Hehre, R. F. Stewart and J. A. Pople, "Self-consistent molecular-orbital methods. I. Use of Gaussian expansions of Slater-type atomic orbitals," *J. Chem. Phys.* **51**, 2657 (1969).

[17] W. Kutzelnigg, "Theory of the expansion of wave functions in a Gaussian basis," *Int. J. Quantum Chem.* **51**, 447 (1994).

[18] D. Braess, "Asymptotics for the approximation of wave functions by exponential sums," *J. Approx. Theory* **83**, 93 (1995).

[19] E. J. Weniger, "The spherical tensor gradient operator," arXiv:math-ph/0505018 (2005).

[20] D. G. A. Smith *et al.*, "PSI4 1.4: open-source software for high-throughput quantum chemistry," *J. Chem. Phys.* **152**, 184108 (2020); the fit is in `psi4/src/psi4/libmints/integralparameters.cc`, class `FittedSlaterCorrelationFactor`.

[21] K. Pachucki, M. Puchalski and E. Remiddi, "Recursion relations for Hylleraas three-electron integral," *Phys. Rev. A* **70**, 032502 (2004); arXiv:physics/0405057.

[22] D. H. Bailey, J. M. Borwein, D. Broadhurst, and M. L. Glasser, "Elliptic integral evaluations of Bessel moments and applications," *J. Phys. A* **41**, 205203 (2008); arXiv:0801.0891.

[23] W. N. Bailey, "Some infinite integrals involving Bessel functions (II)," J. London Math. Soc. **11**, 16–20 (1936).

[24] F. C. S. Brown, "On the periods of some Feynman integrals," arXiv:0910.0114 (2009).

[25] S. Bloch and P. Vanhove, "The elliptic dilogarithm for the sunset graph," *J. Number Theory* **148**, 328–364 (2015); arXiv:1309.5865.

[26] J. Fresán, C. Sabbah, and J.-D. Yu, "Quadratic relations between Bessel moments," *Algebra Number Theory* **17**, 541–602 (2023); arXiv:2006.02702.

[27] J. E. Avery, "Fast electron repulsion integrals for molecular Coulomb Sturmians," *Adv. Quantum Chem.* **67**, 129–151 (2013).

[28] J. E. Avery and J. S. Avery, "Molecular integrals for exponential-type orbitals using hyperspherical harmonics," *Adv. Quantum Chem.* **70**, 265–324 (2015).

[29] G. Almkvist, C. van Enckevort, D. van Straten, and W. Zudilin, "Tables of Calabi–Yau equations," arXiv:math/0507430 (2005).

[30] Calabi–Yau differential operator database, version 3.0, `cydb.mathematik.uni-mainz.de`, entry 2.9 (AESZ #58).

[31] N. Gegelia and D. van Straten, "Paramodular forms from Calabi–Yau operators," arXiv:2408.10183 (2024), Tables 3 and 6.

[32] J. Matuzas, "A Calabi–Yau period in a four-electron correlation integral: modular physical fibre and a hypergeometric endpoint," Zenodo, doi:10.5281/zenodo.23218203 (2026).

[33] G. Almkvist and W. Zudilin, "Differential equations, mirror maps and zeta values," in *Mirror Symmetry V*, AMS/IP Studies in Advanced Mathematics **38**, 481–515 (2007); arXiv:math/0402386.

[34] P. Vanhove, "The physics and the mixed Hodge structure of Feynman integrals," arXiv:1401.6438 (2014).

[35] F. Beukers and G. Heckman, "Monodromy for the hypergeometric function ${}_nF_{n-1}$," *Invent. Math.* **95**, 325–354 (1989).

[36] Z. L. Lim, F.-T. Tu and J.-D. Yu, "Galois representations and modularity for twisted moments of Kloosterman sums," arXiv:2609.13576 (2026).

[37] A. I. Davydychev and R. Delbourgo, "A geometrical angle on Feynman integrals," J. Math. Phys. **39**, 4299 (1998), arXiv:hep-th/9709216.

[38] B. G. Nickel, J. Math. Phys. **19**, 542 (1978).

[39] M. Preti, "STR: a Mathematica package for the method of uniqueness," arXiv:1811.04935 (2018).

[40] D. Broadhurst, "Bessel moments, random walks and Calabi–Yau equations," preprint (2009).

[41] J. L. Bourjaily, A. J. McLeod, M. von Hippel, and M. Wilhelm, "Bounded collection of Feynman integral Calabi–Yau geometries," *Phys. Rev. Lett.* **122**, 031601 (2019); arXiv:1810.07689.

[42] B. Padhy, "Analytic evaluation of some three- and four-electron atomic integrals involving s STO's and exponential correlation with unlinked r_{ij}'s," arXiv:1701.05701 (2017).

[43] K. Bönisch, F. Fischbach, A. Klemm, C. Nega, and R. Safari, "Analytic structure of all loop banana integrals," *J. High Energy Phys.* **05**, 066 (2021); arXiv:2008.10574.

[44] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," Paper 18 in this series.

[45] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," Paper 34 in this series.

[46] J. Loutey, "The Bessel-Moment Period Algebra of the Three-Center Coulomb Integral: Modular Structure and Cosmic-Galois Placement," Paper 61 in this series.